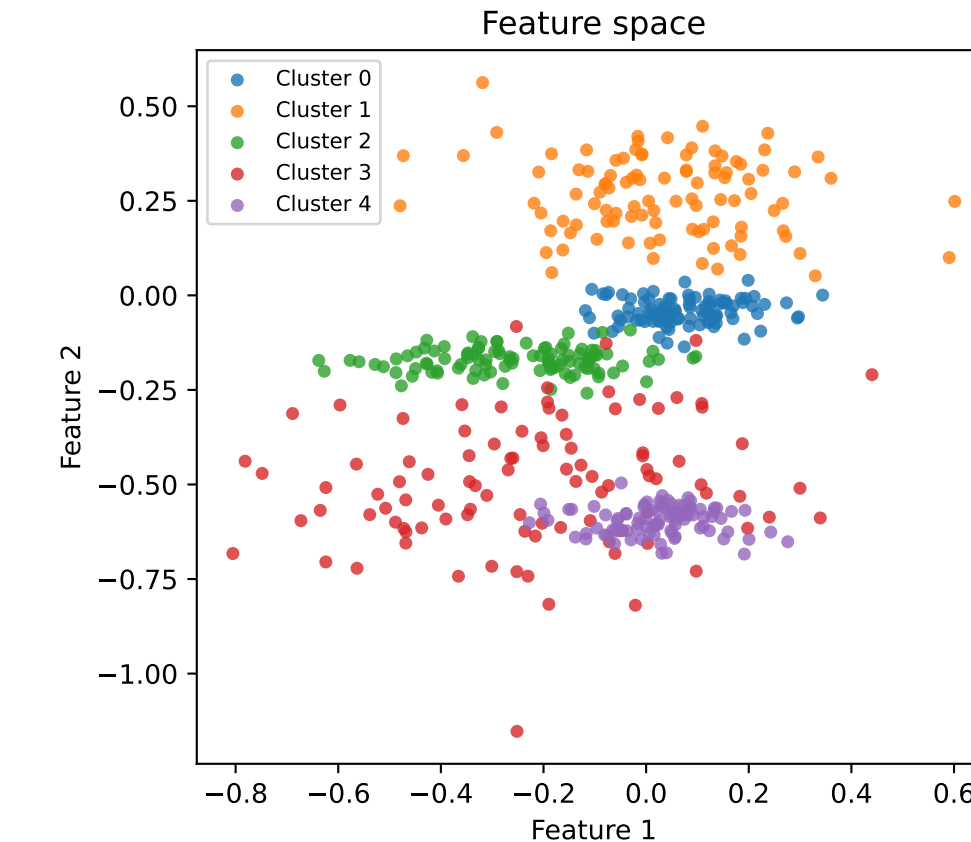
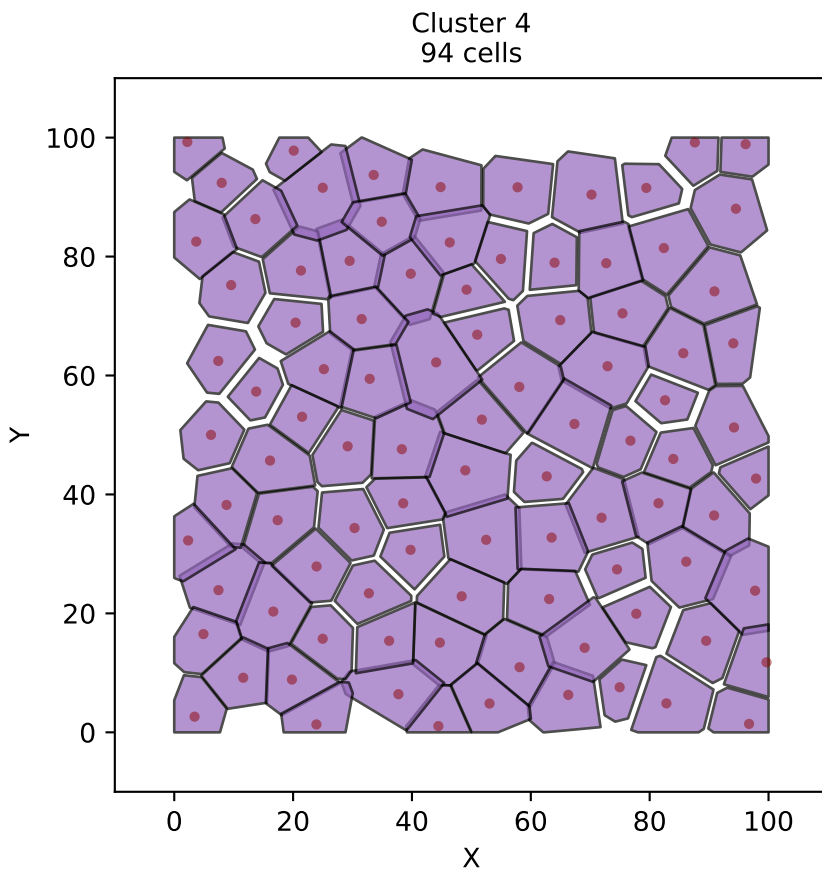
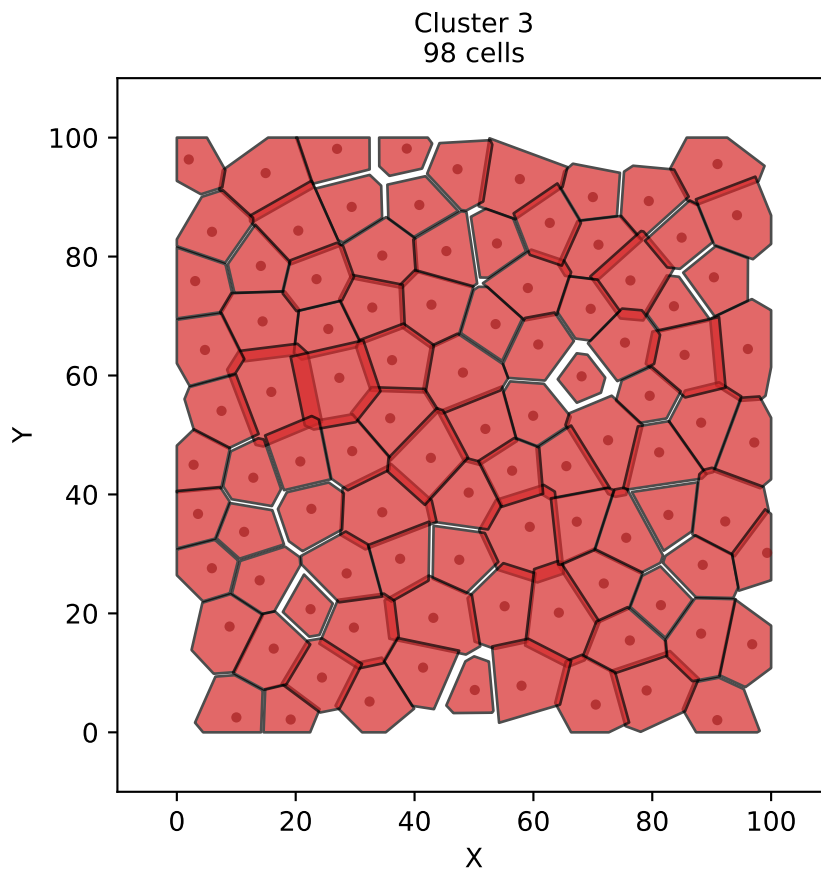
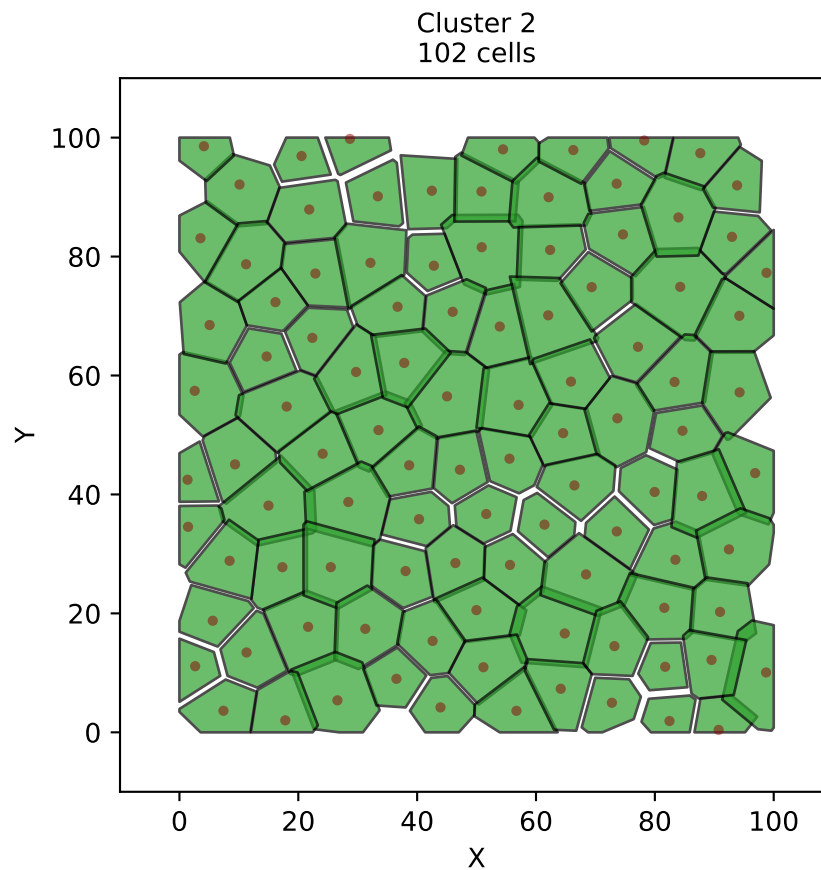
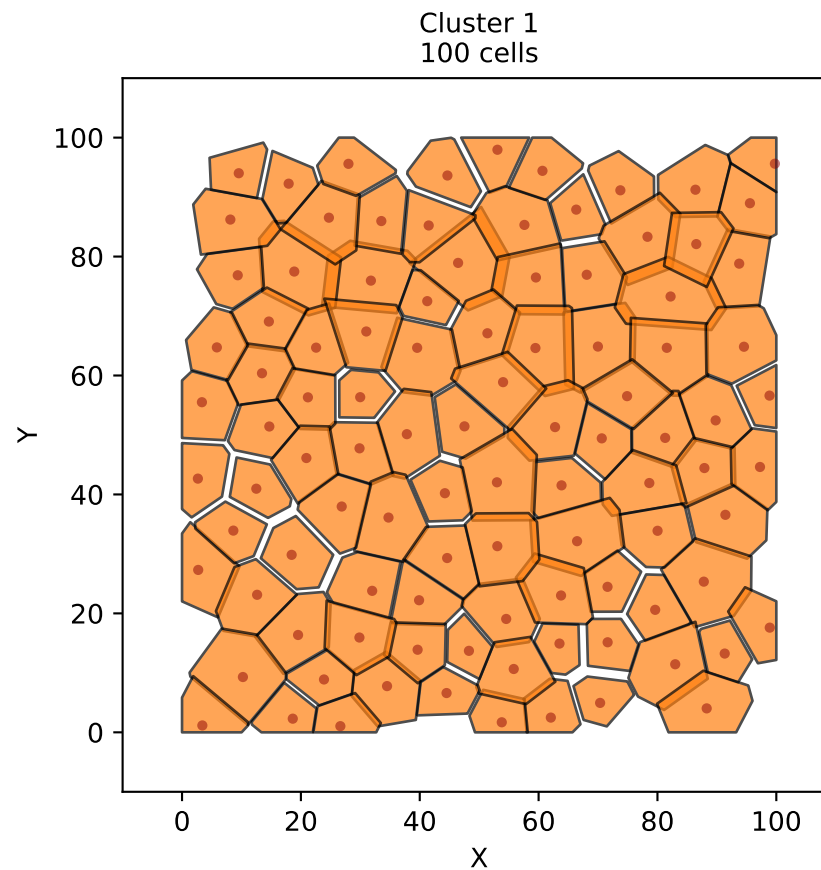
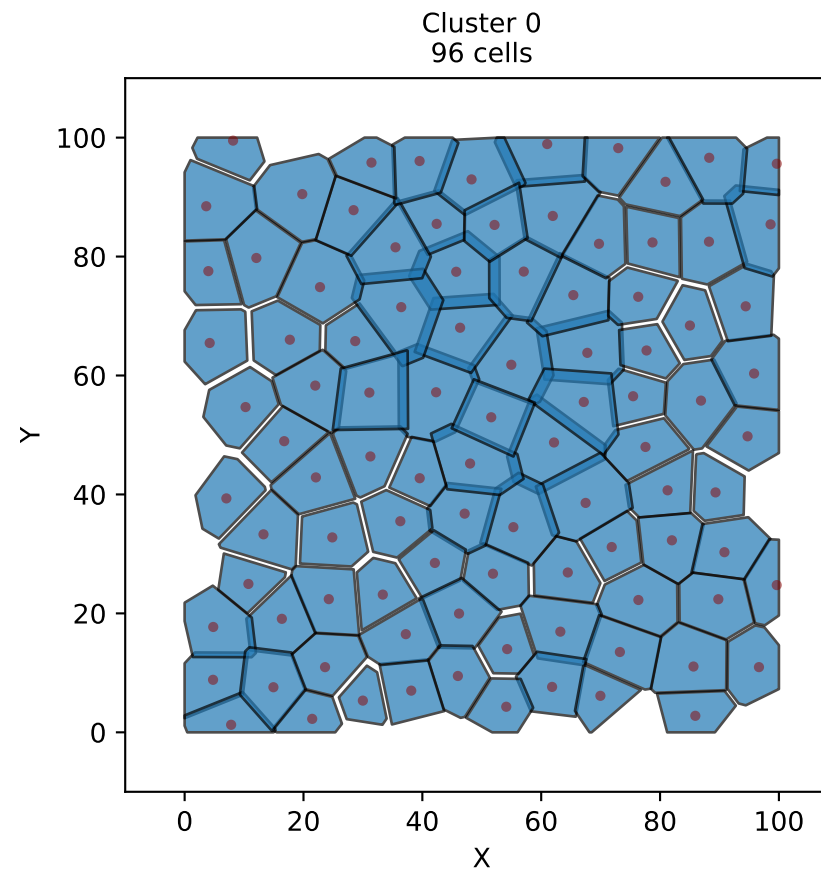
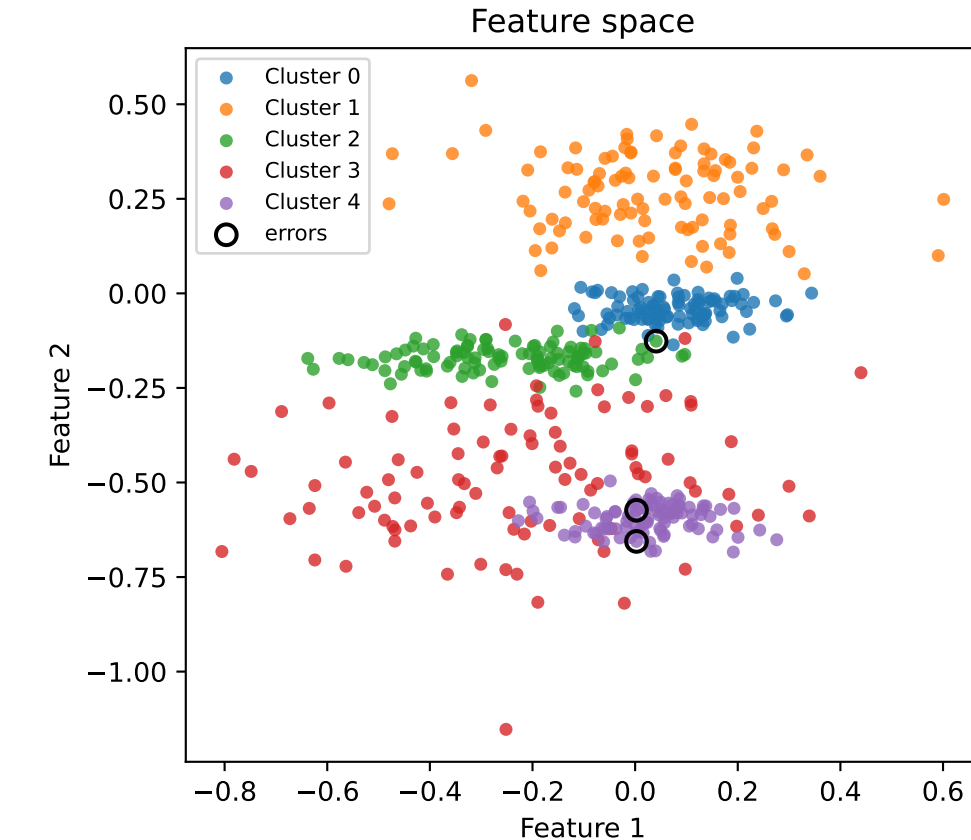
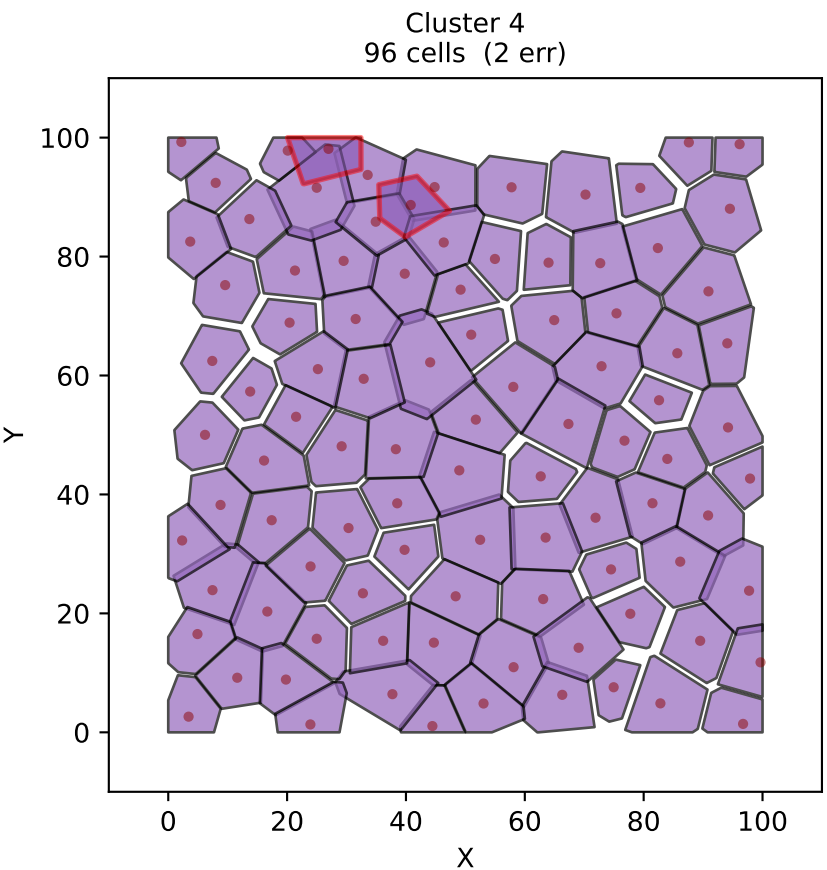
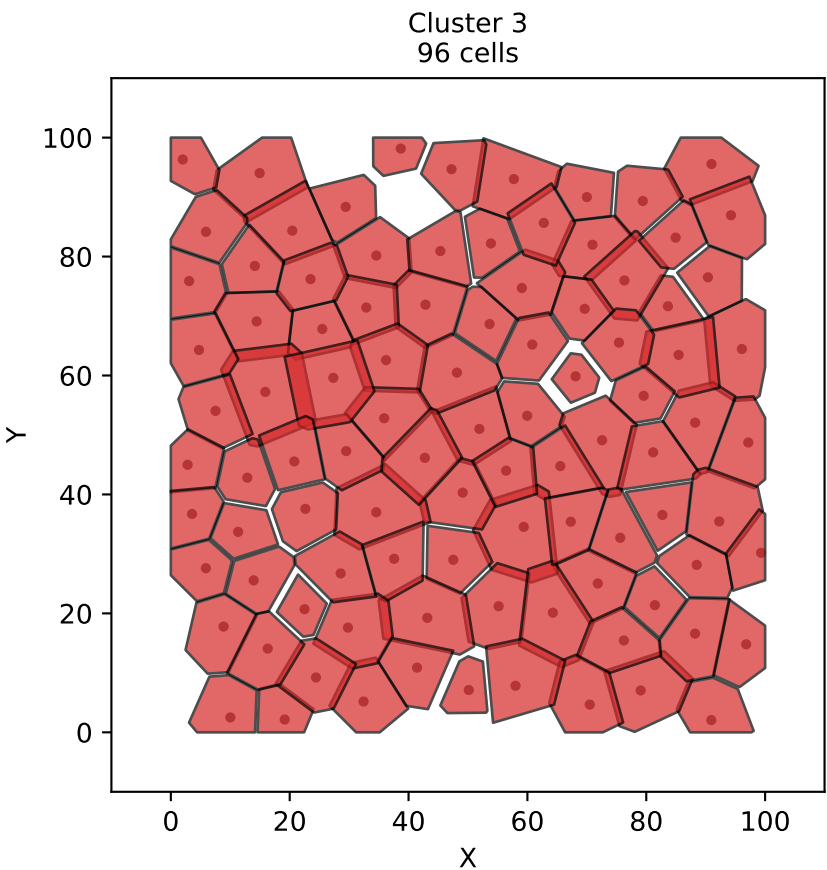
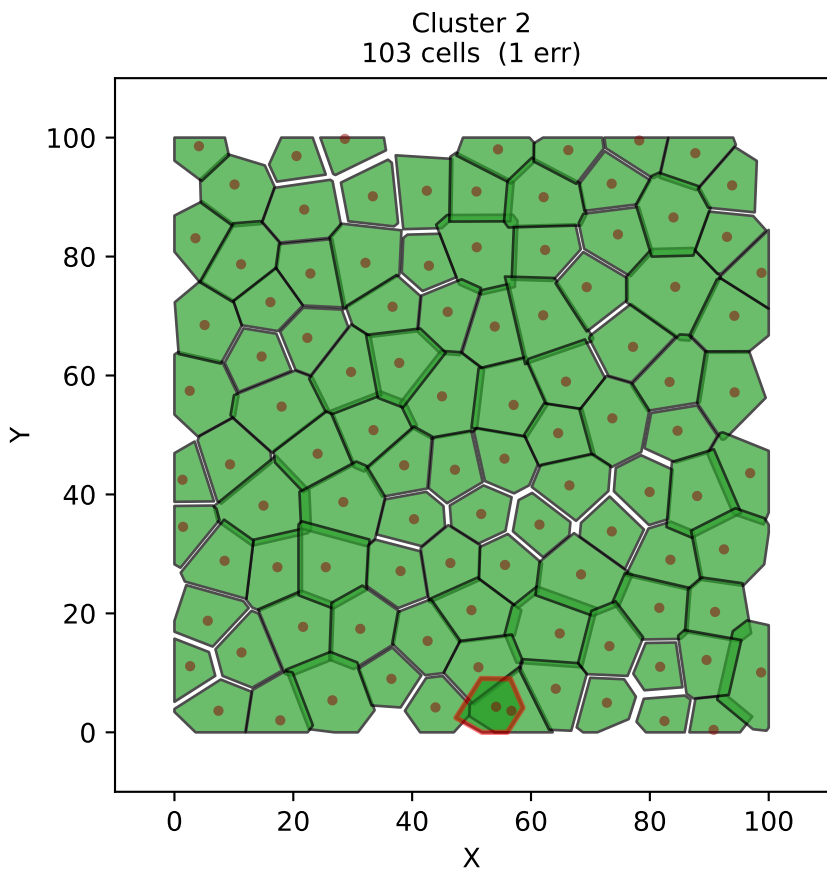
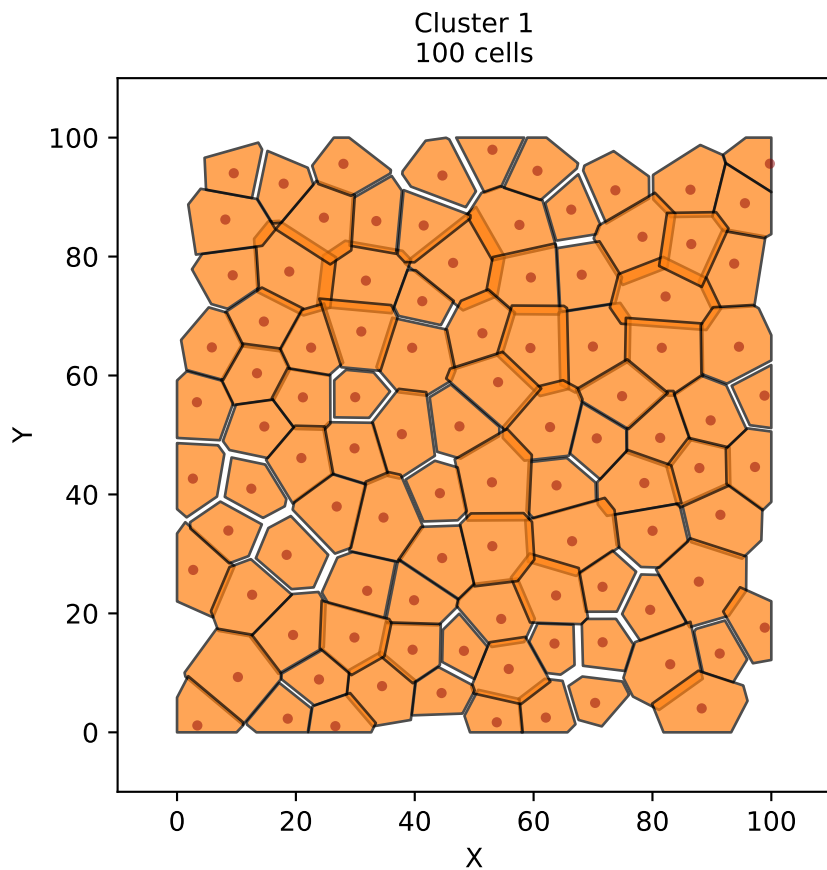
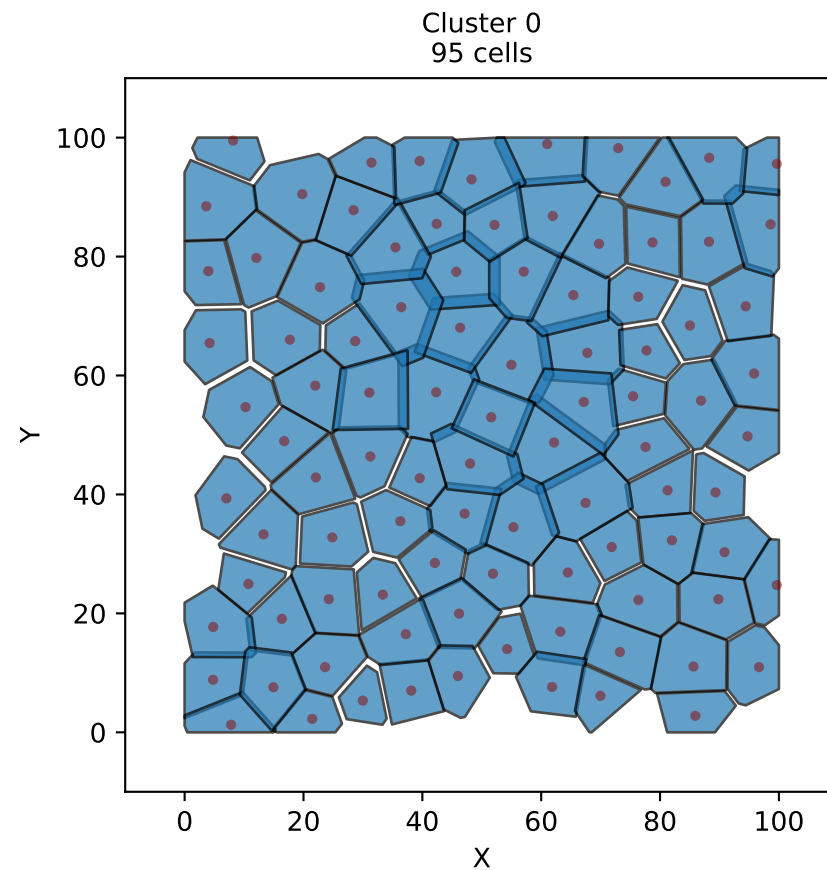


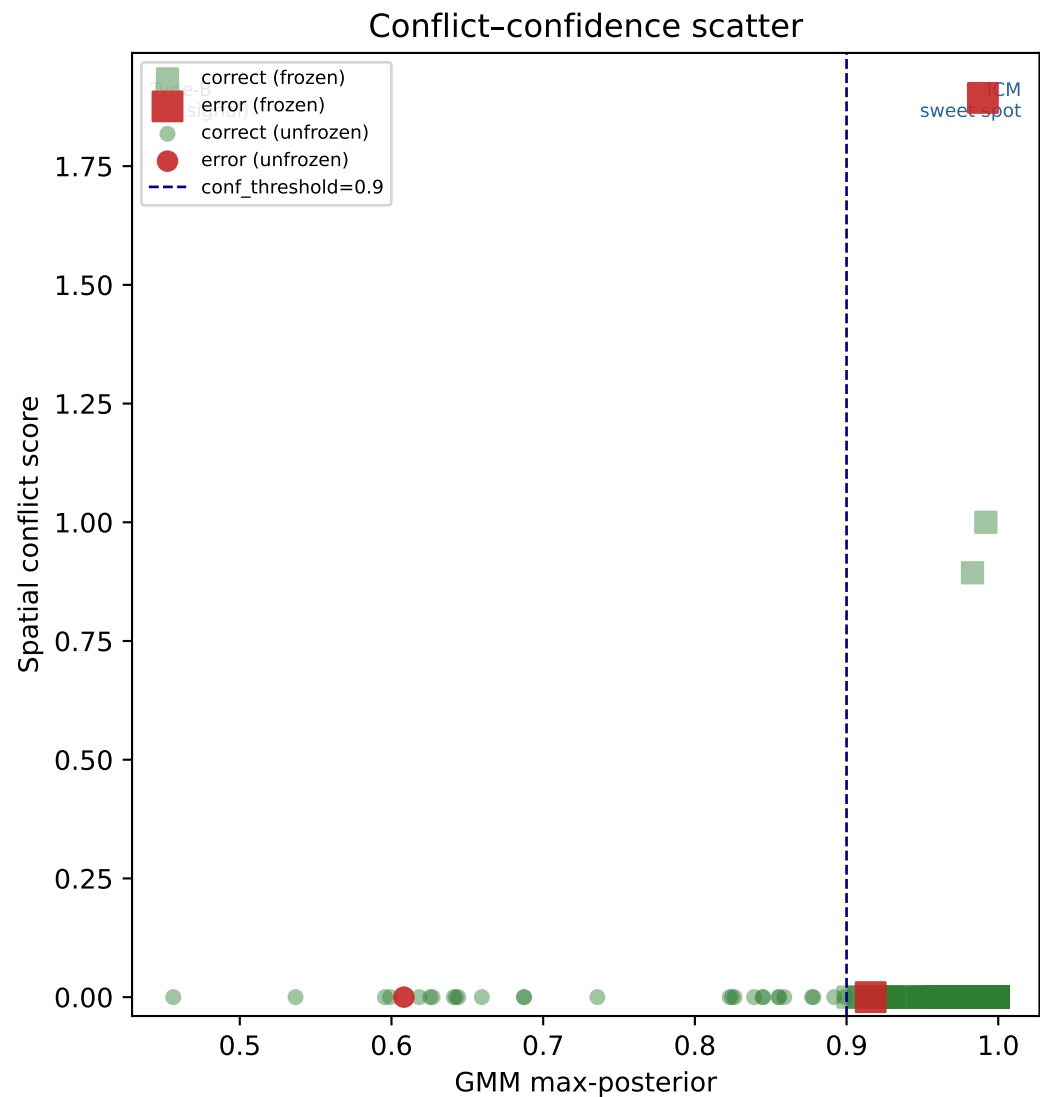
elongated — ground truth



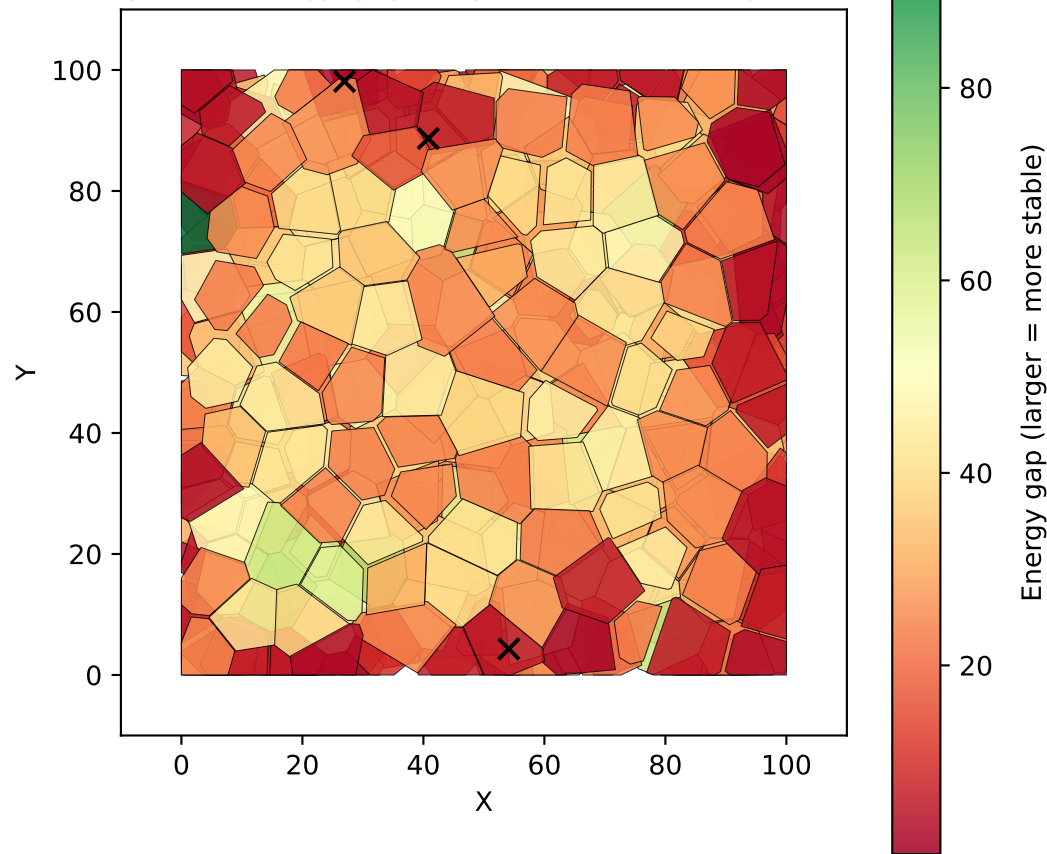
elongated — GMM + MRF ARI=0.985 violations 17 -> 2 changed=9 iters=2



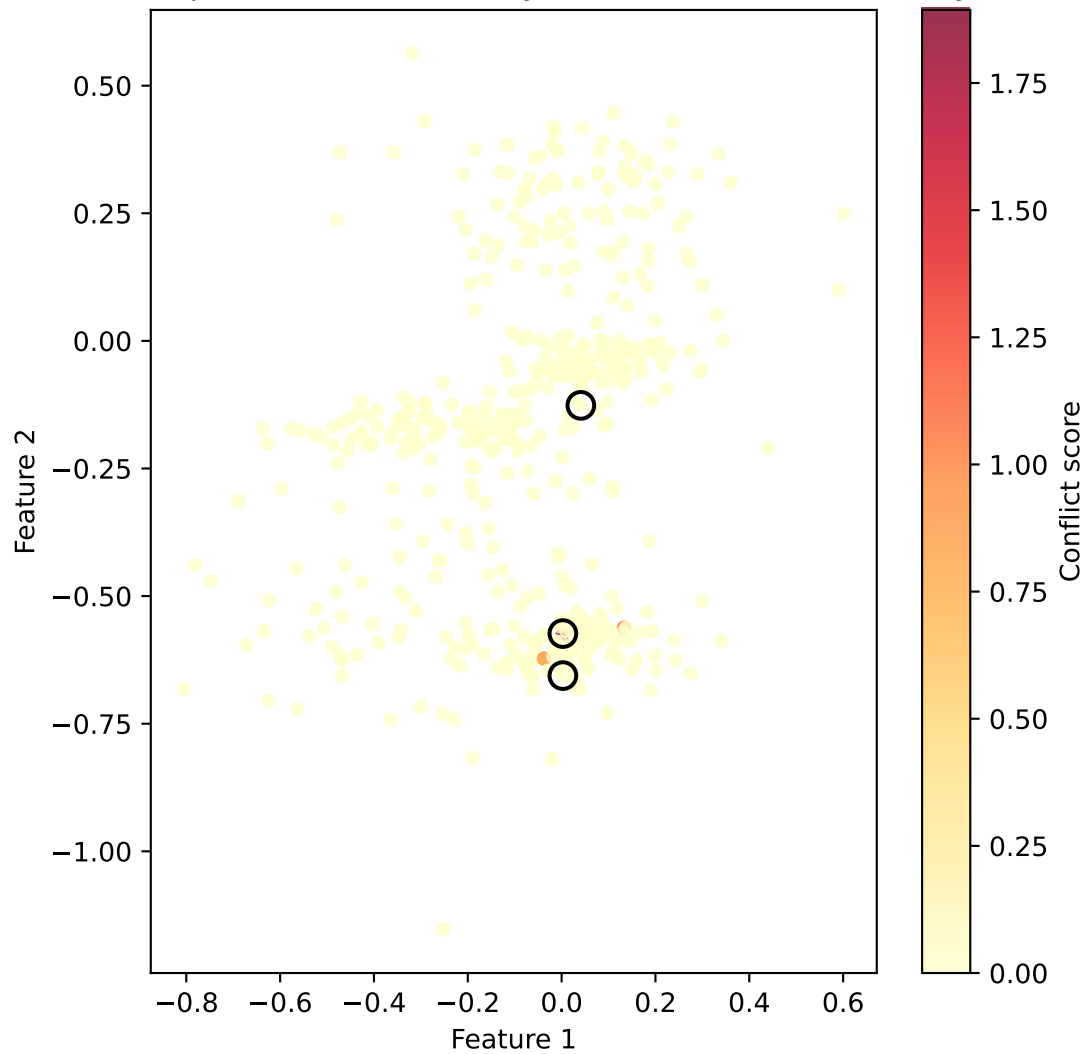
elongated — MRF diagnostics



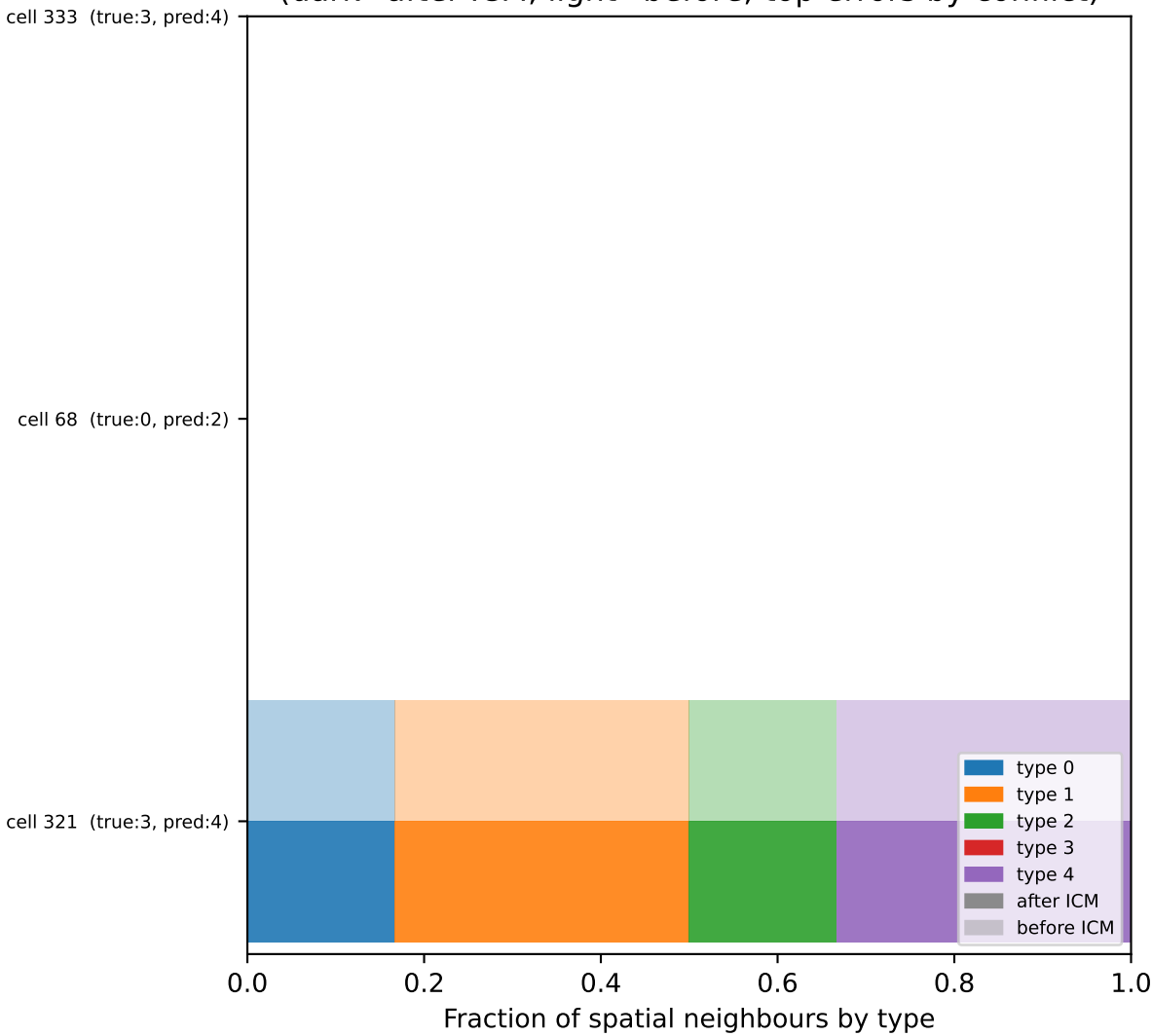
Spatial energy-gap map (x = remaining error)



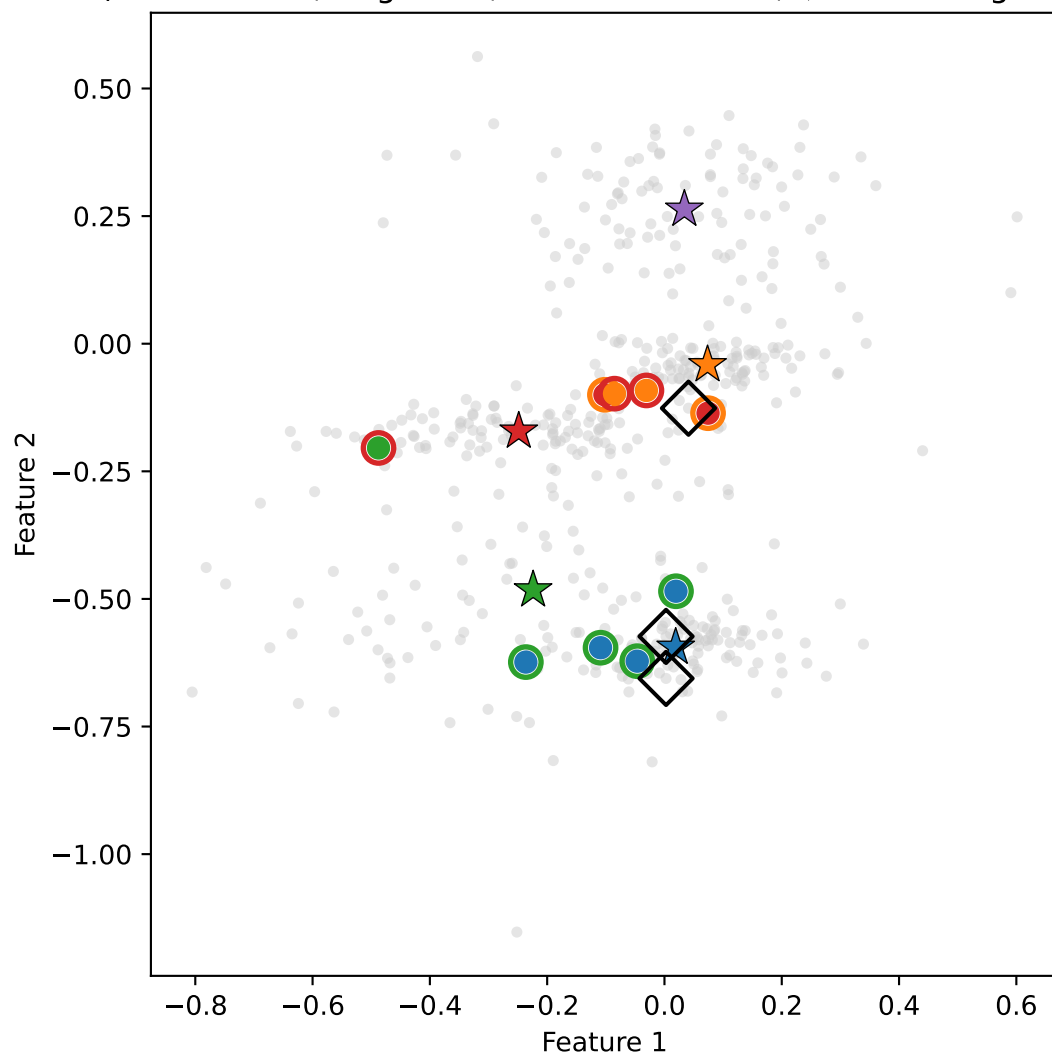
Feature space + conflict (navy = GMM decision boundary)



Neighbour type composition  
(dark=after ICM, light=before; top errors by conflict)



ICM label changes: 9 cells moved  
(fill=old label, ring=new; ★=cluster mean; ◇=remaining error)

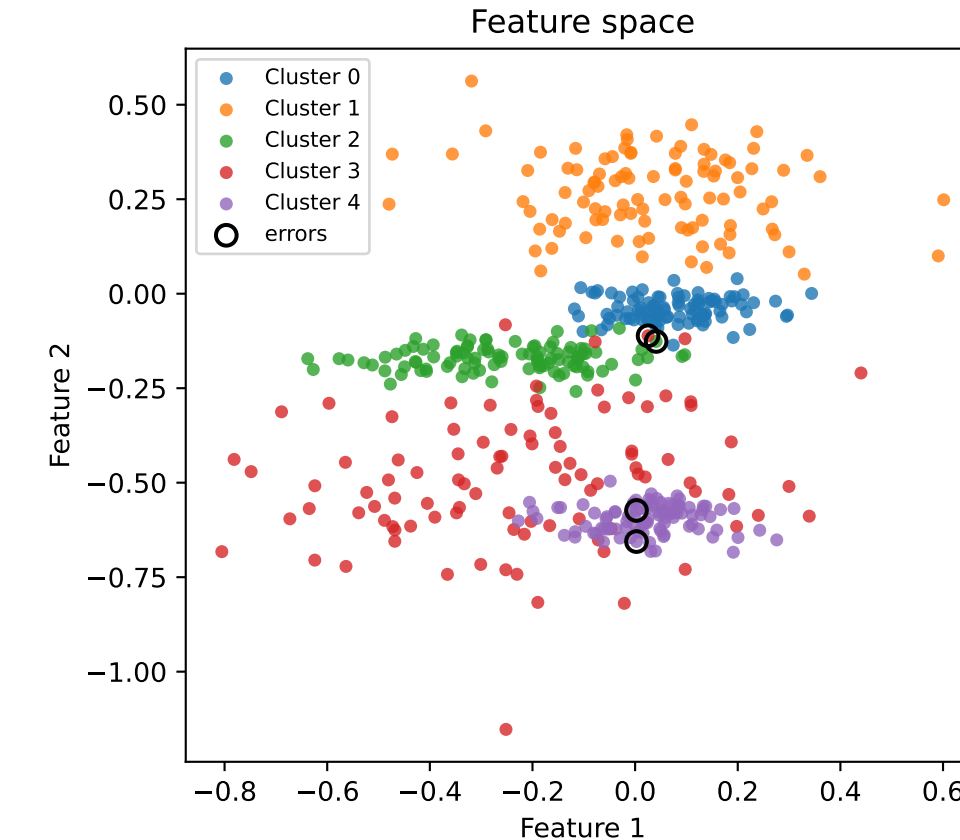
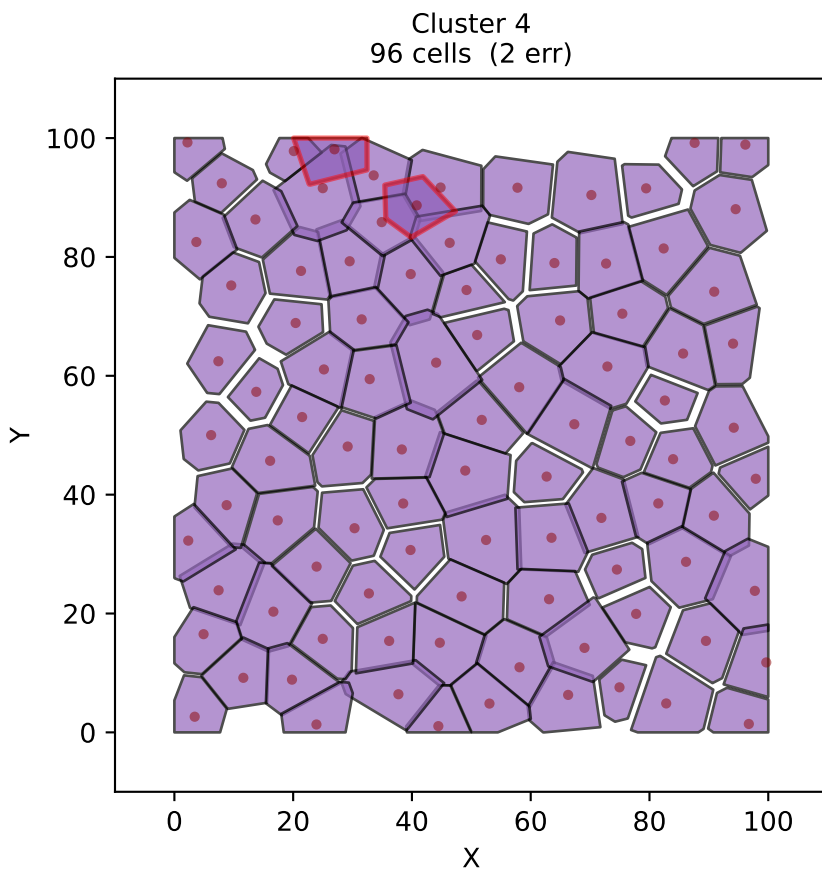
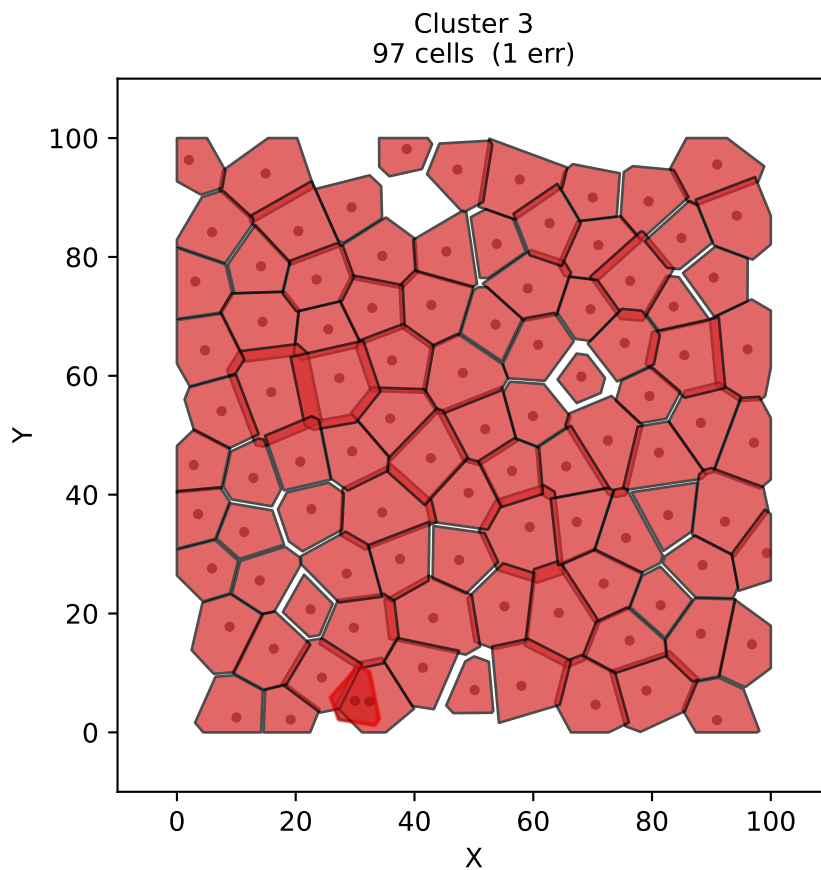
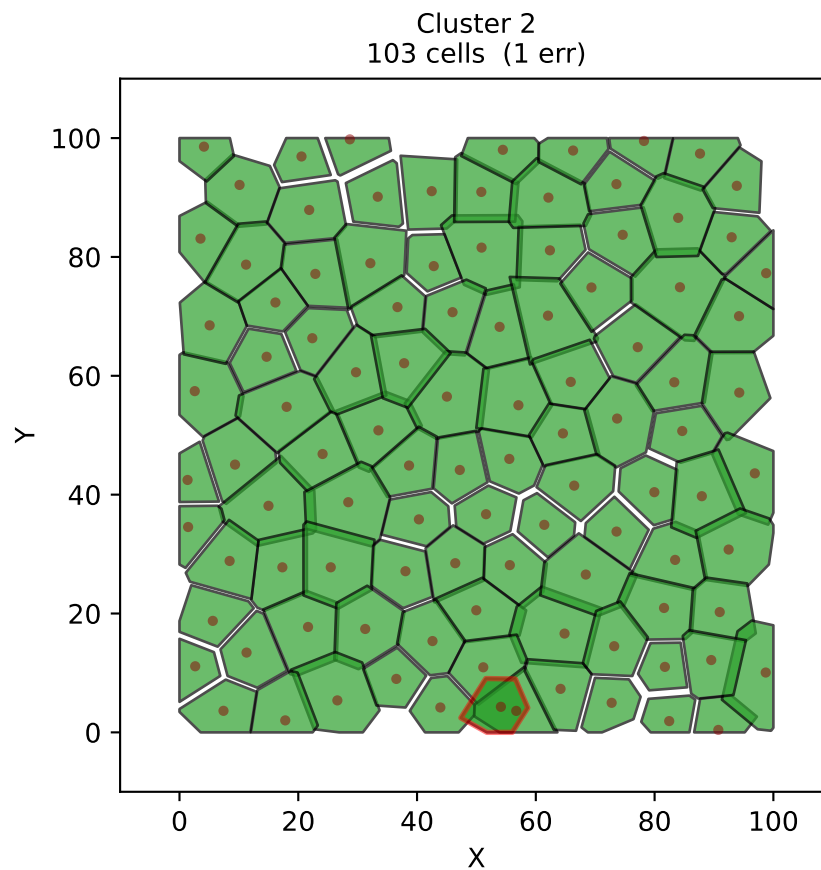
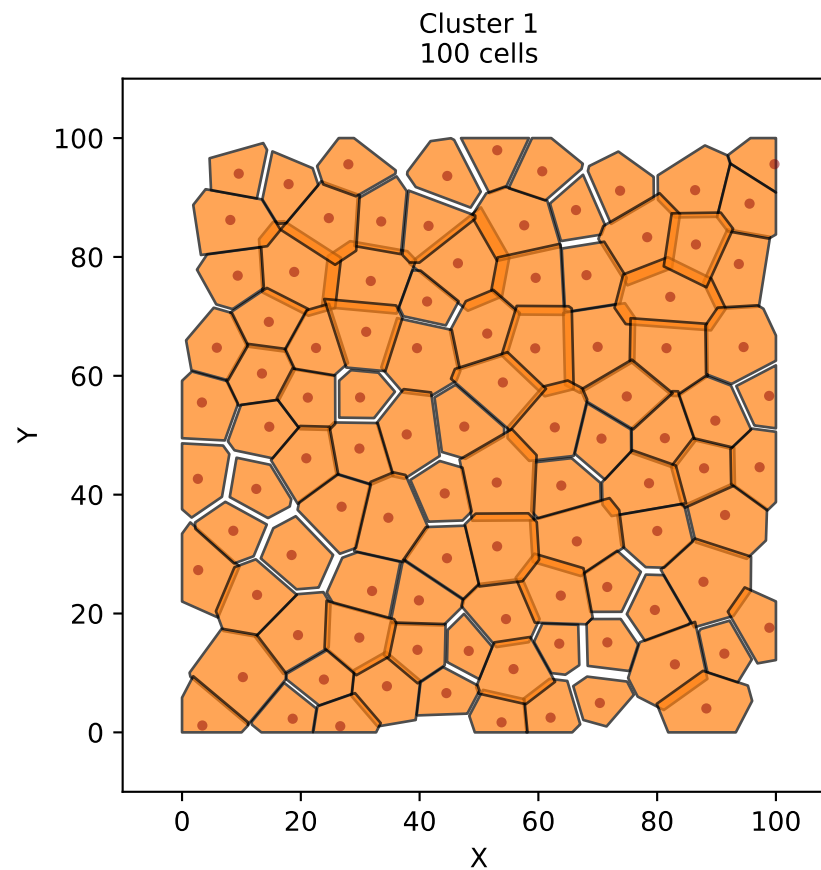
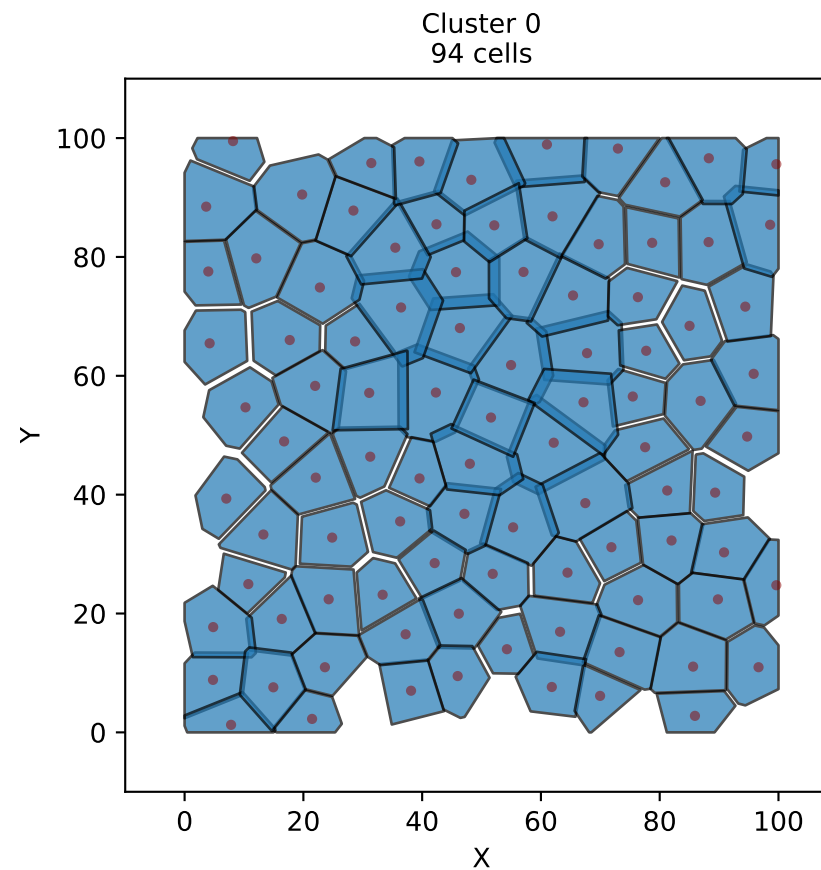


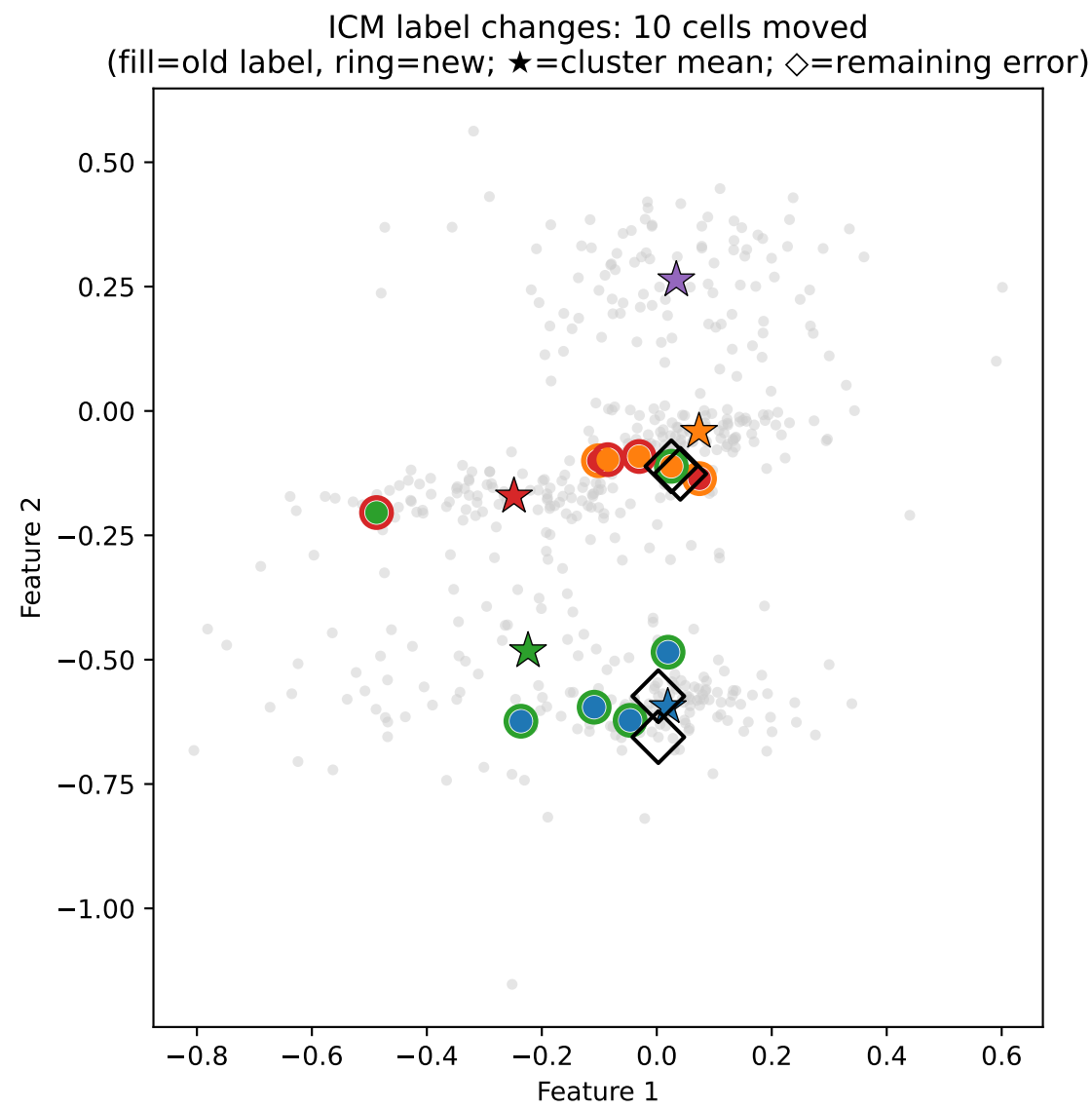
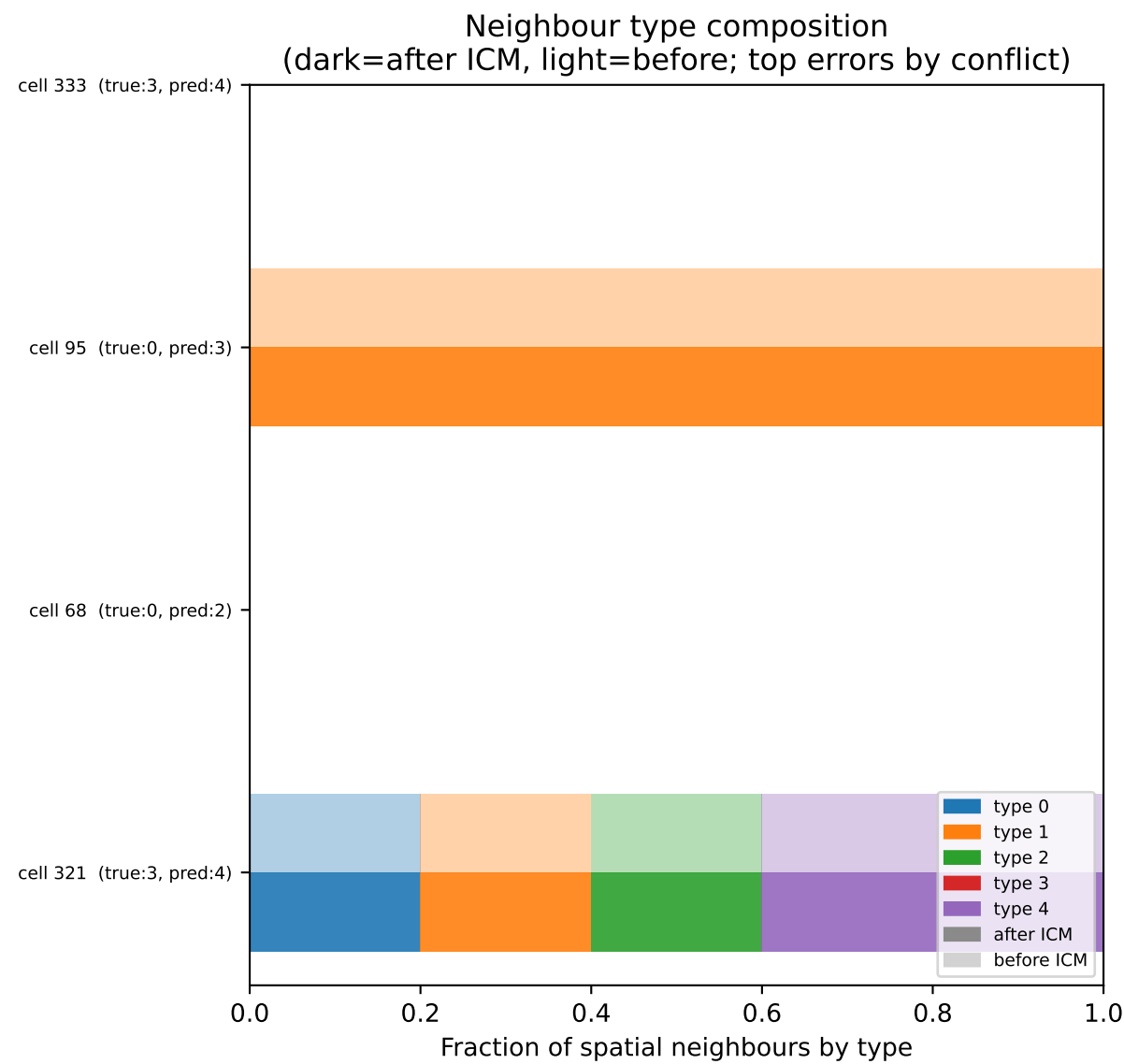
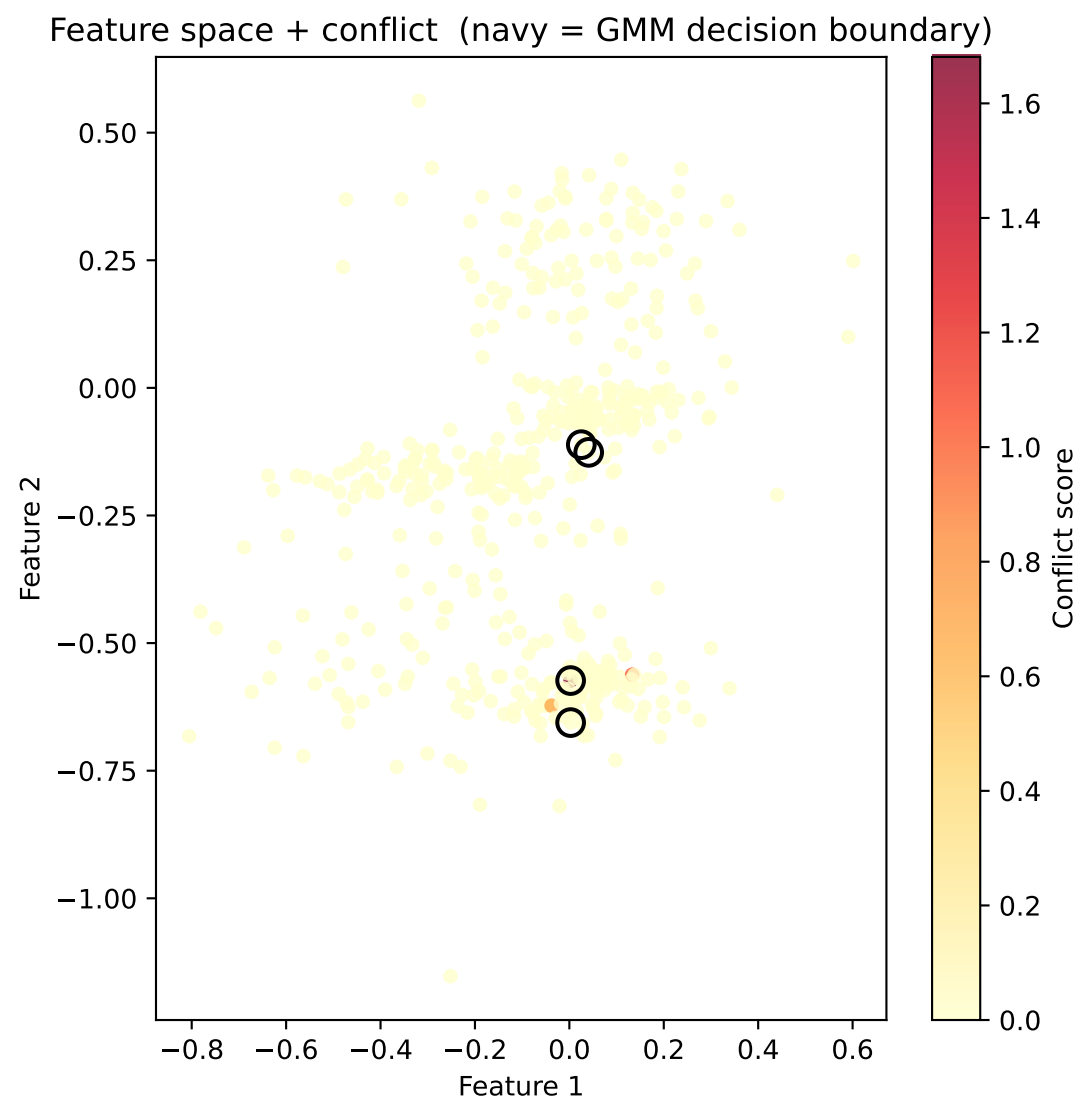
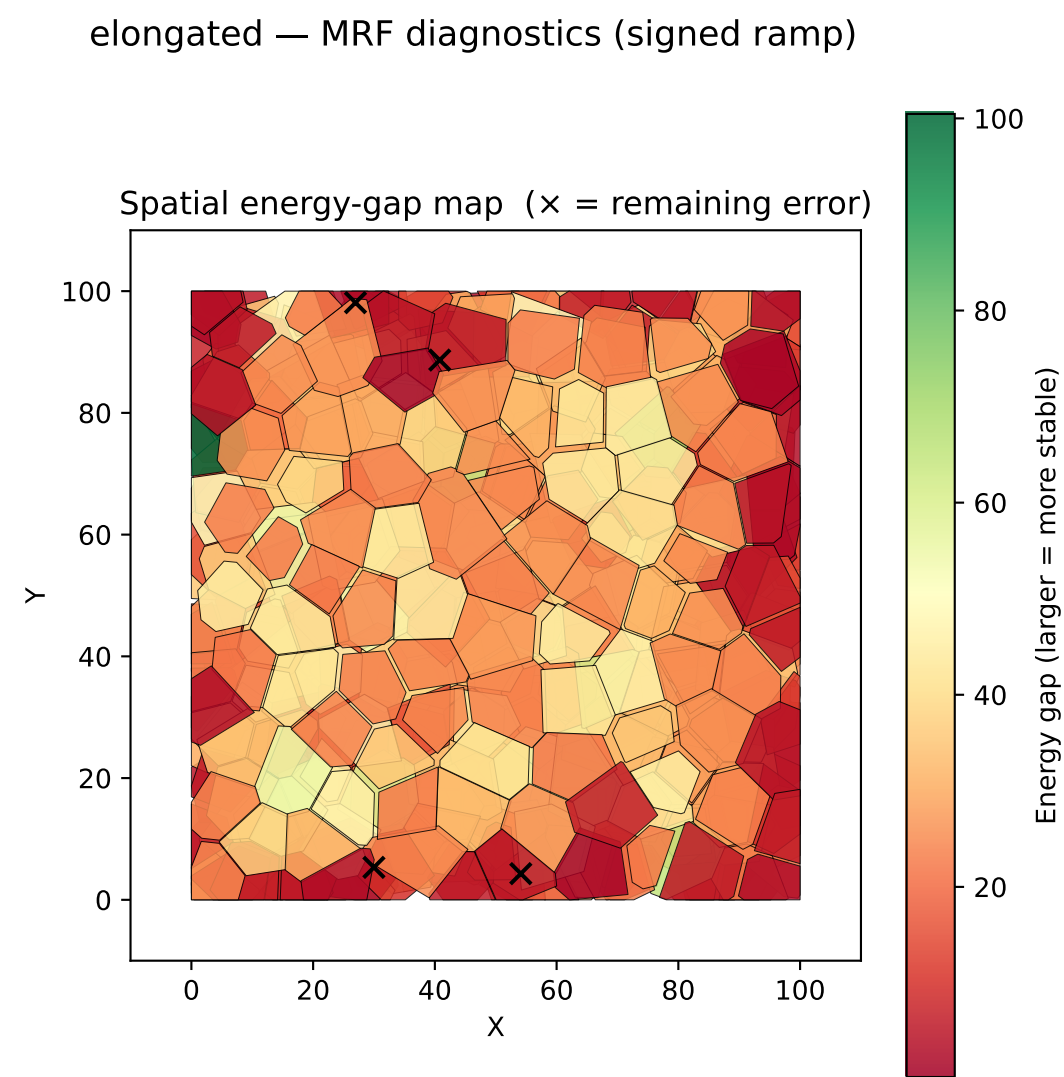
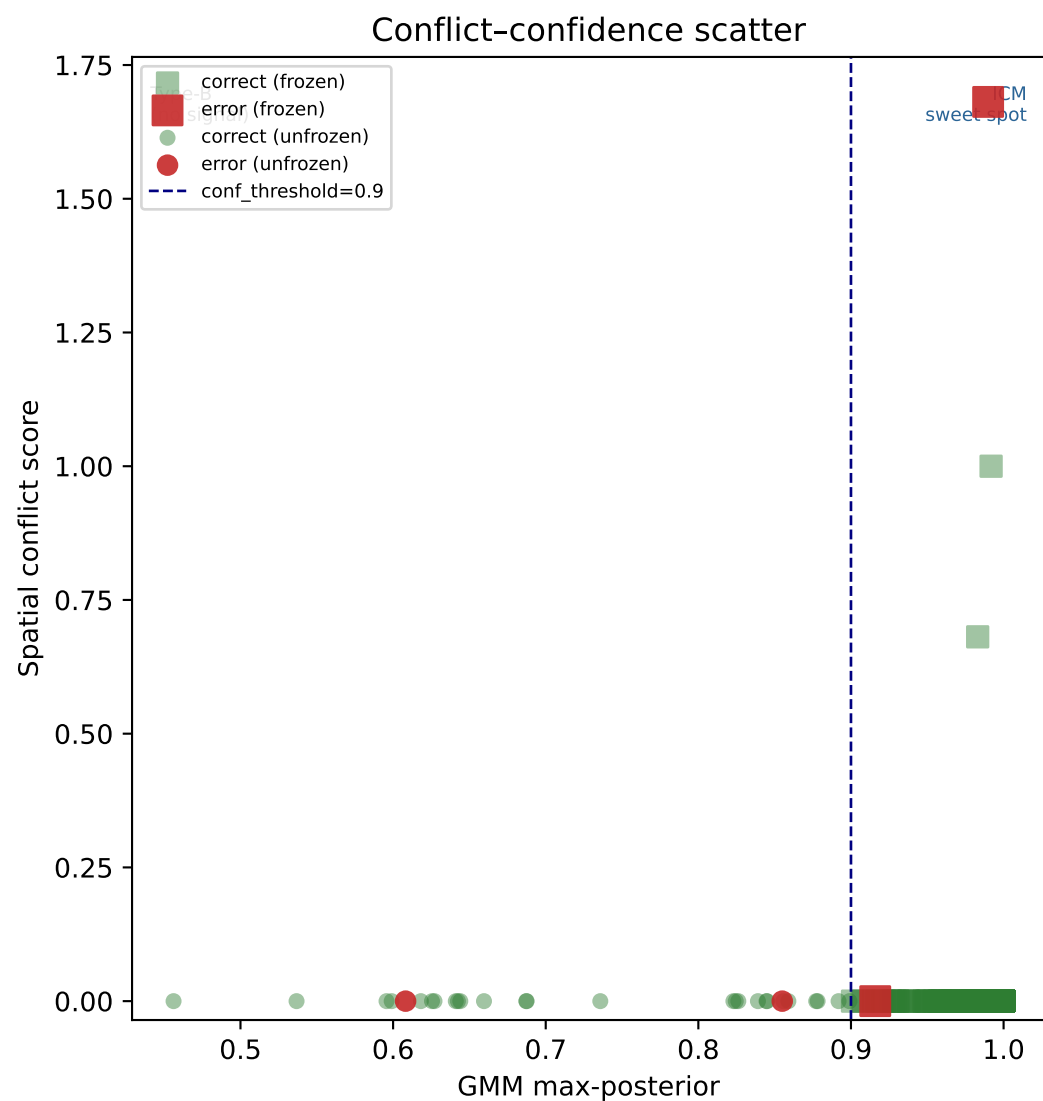
Run summary

| MRF Run Summary |                   |
|-----------------|-------------------|
| ARI:            | 0.9849            |
| errors:         | 3 / 490           |
| violations:     | 17 → 2            |
| n_changed:      | 9                 |
| n_iters:        | 2                 |
| n_frozen:       | 462               |
| Parameters:     |                   |
| lam:            | 20.0              |
| tau_low:        | 0.1               |
| tau_high:       | 0.4               |
| radius:         | 7.059097824976937 |
| conf_thr:       | 0.9               |



elongated — GMM + MRF (signed ramp, tau\_low=0.3) ARI=0.980 violations 16 -> 2 changed=10 iters=2

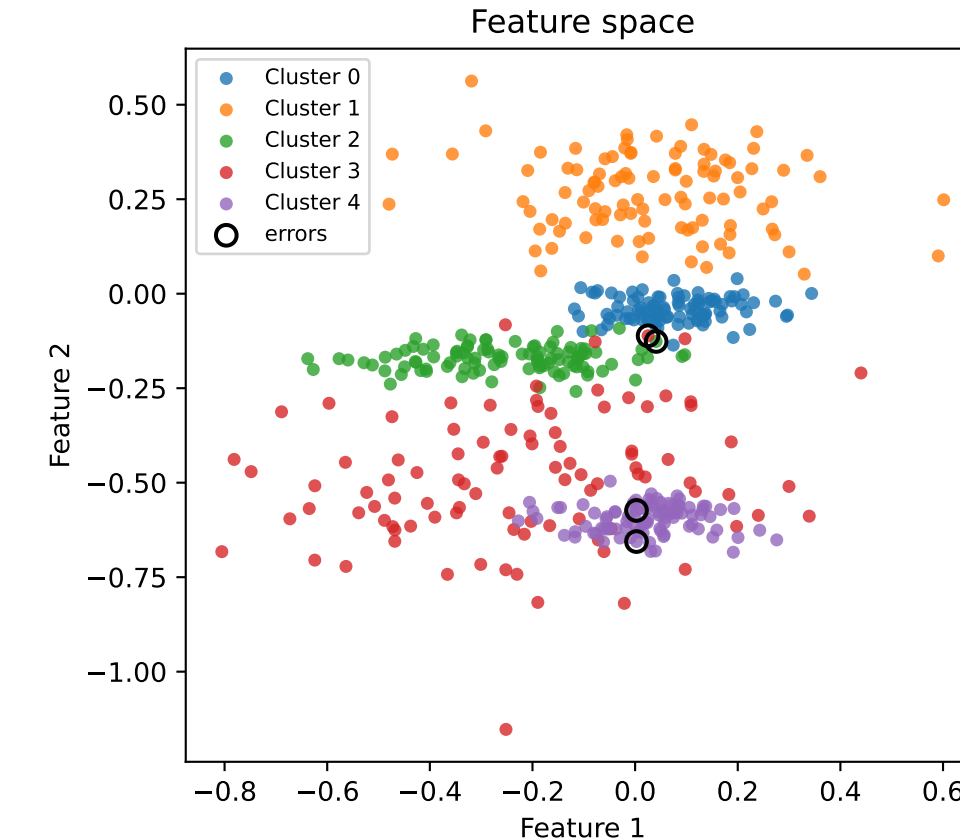
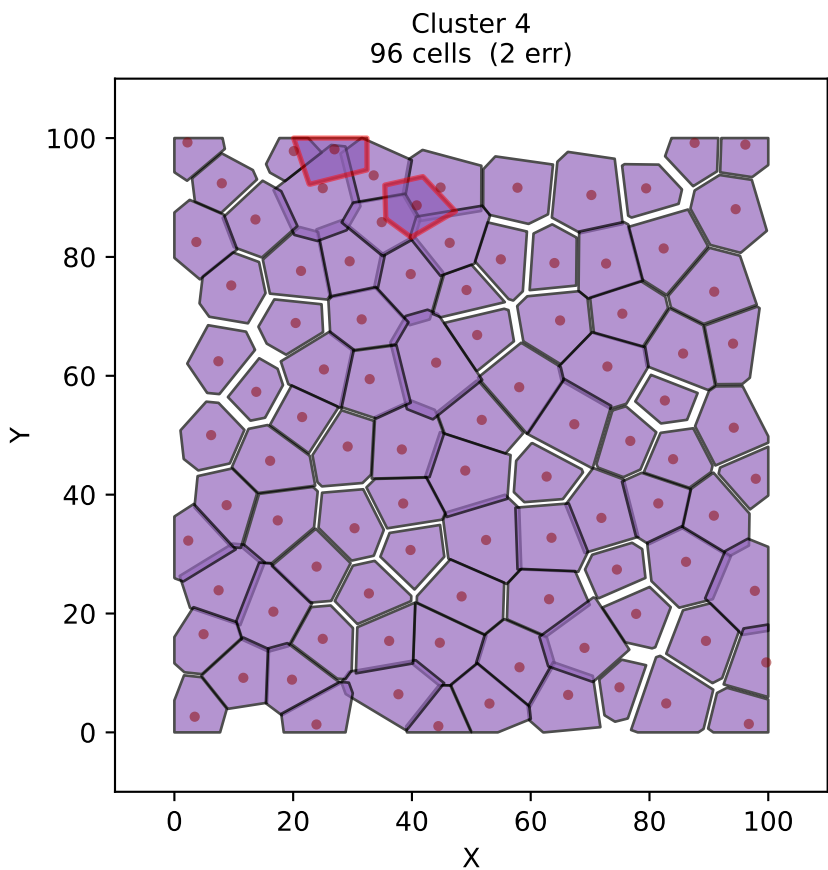
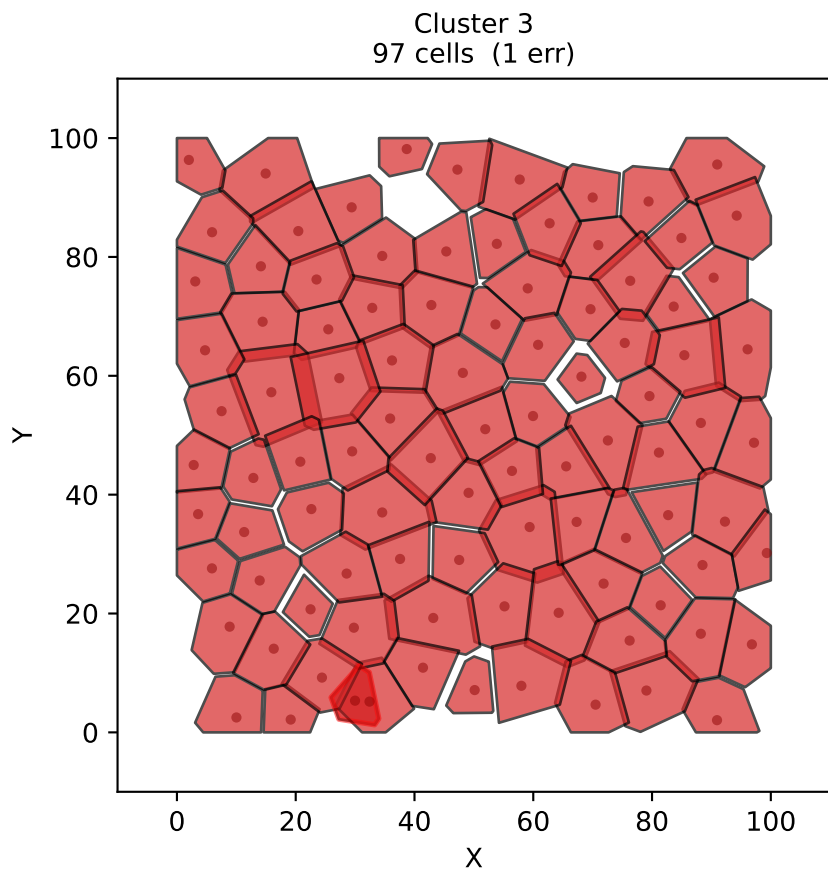
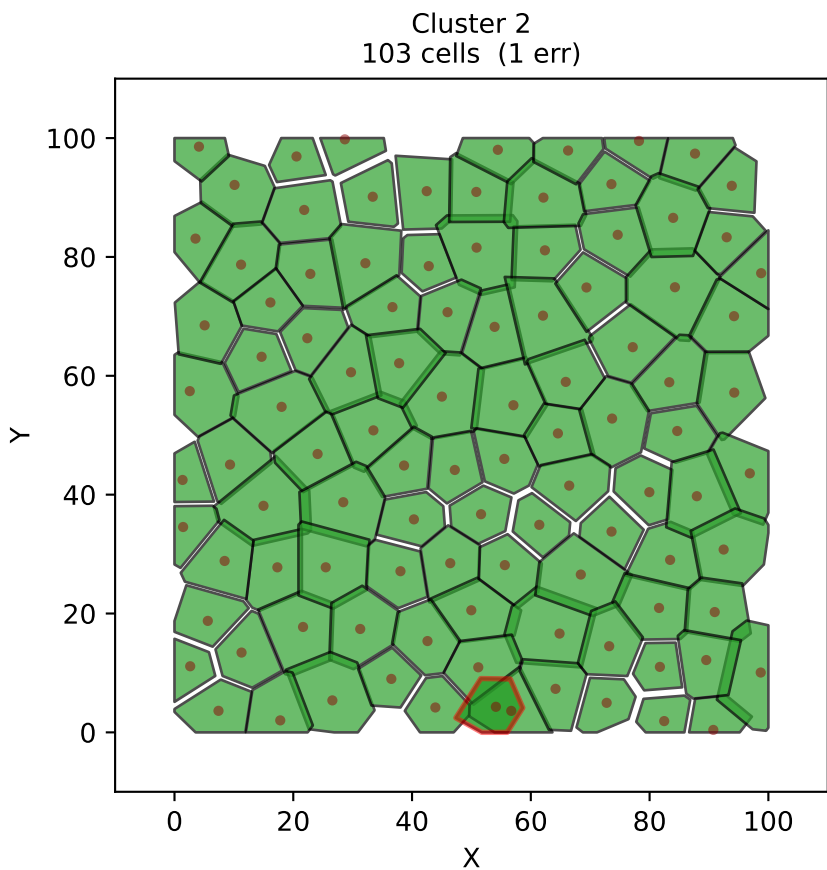
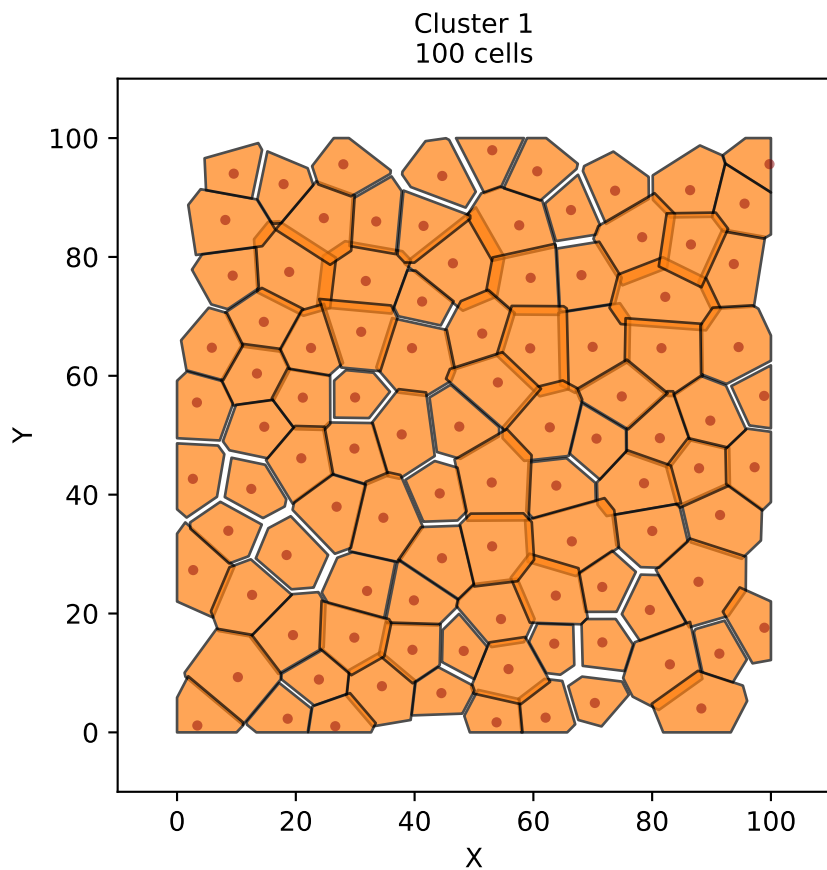
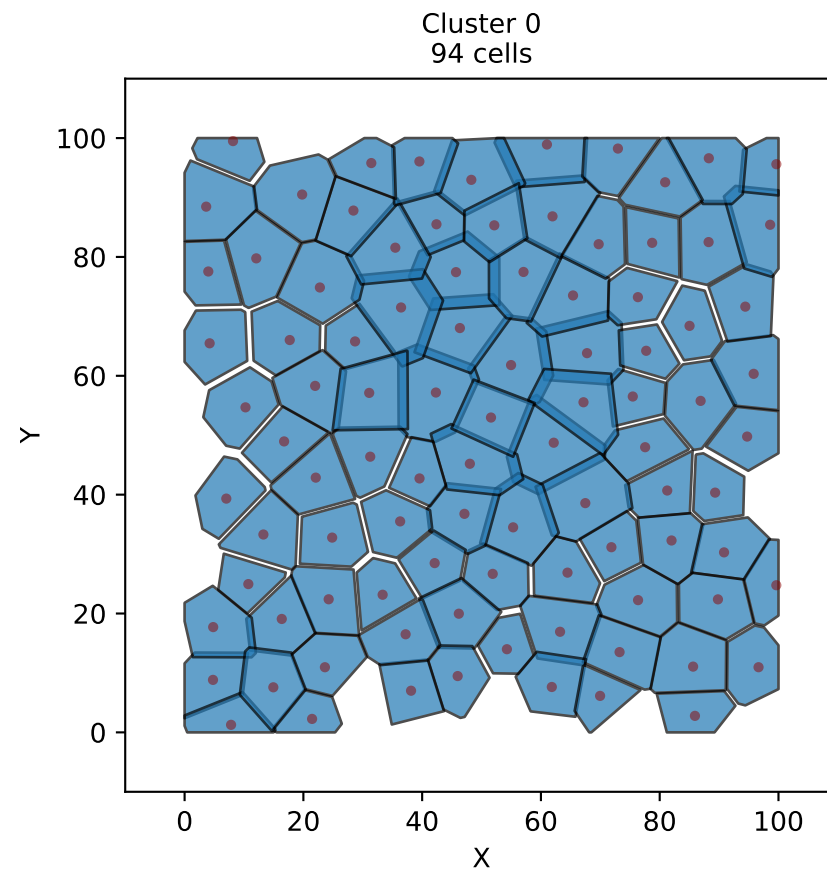




### Run summary

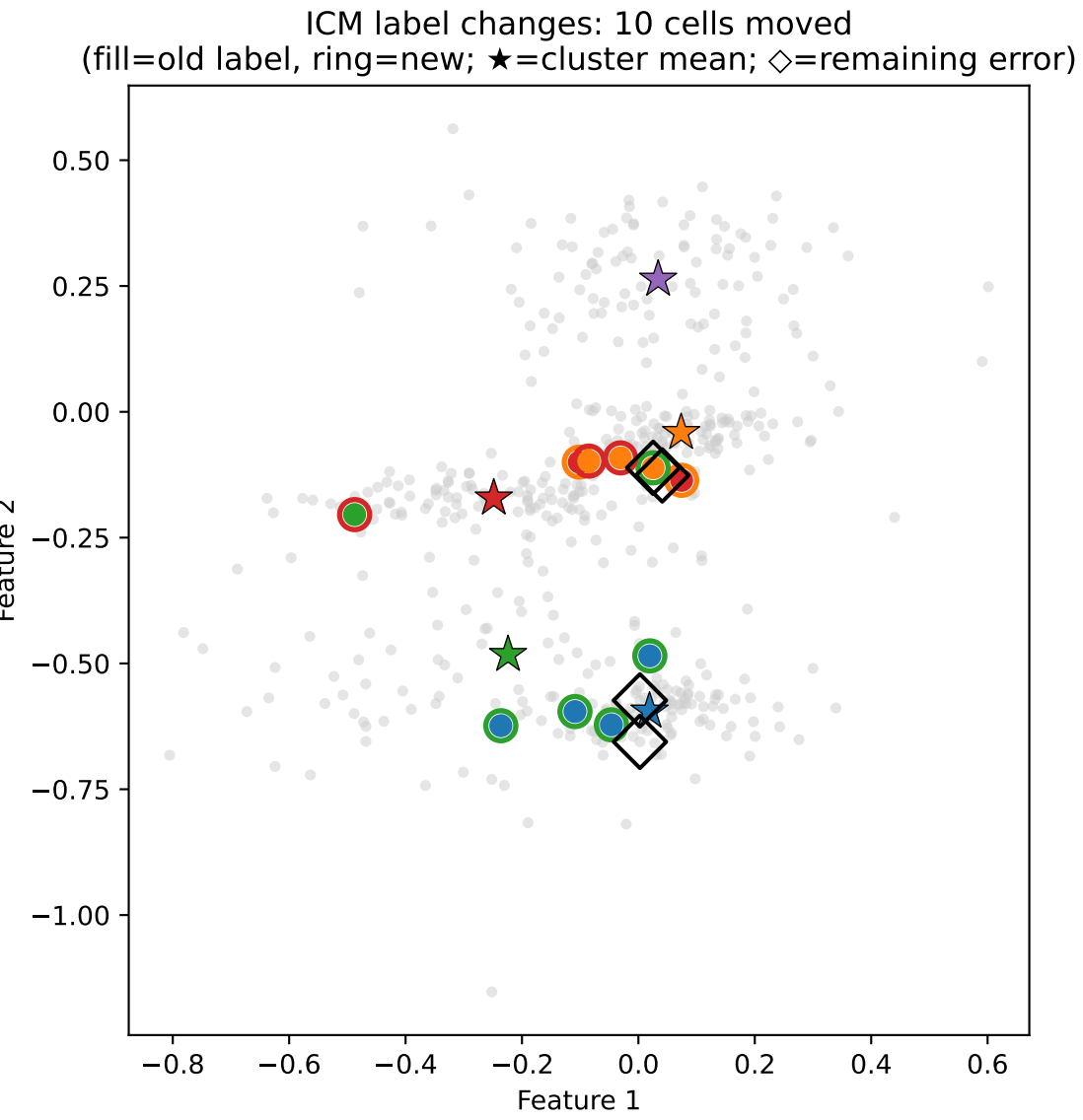
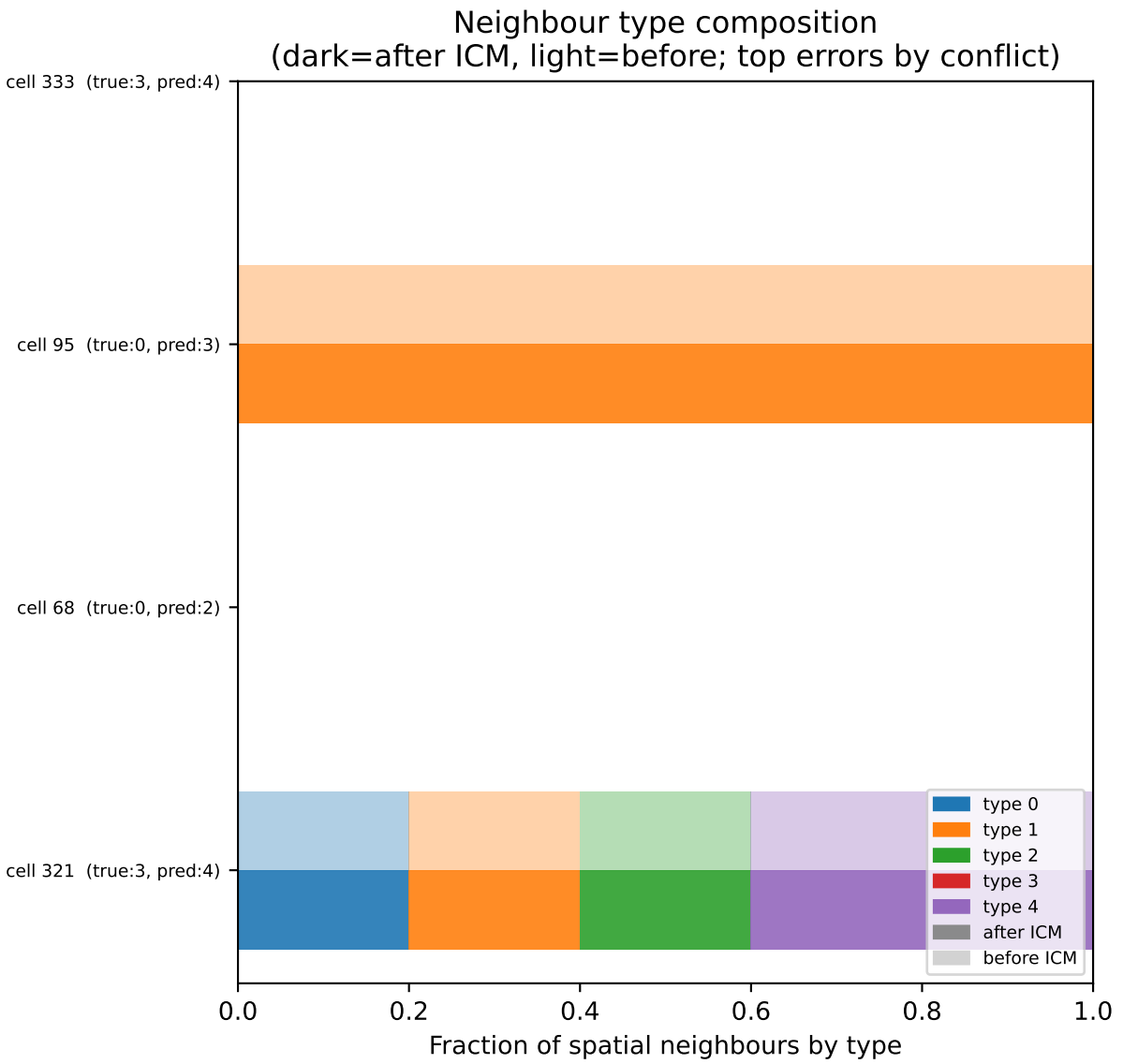
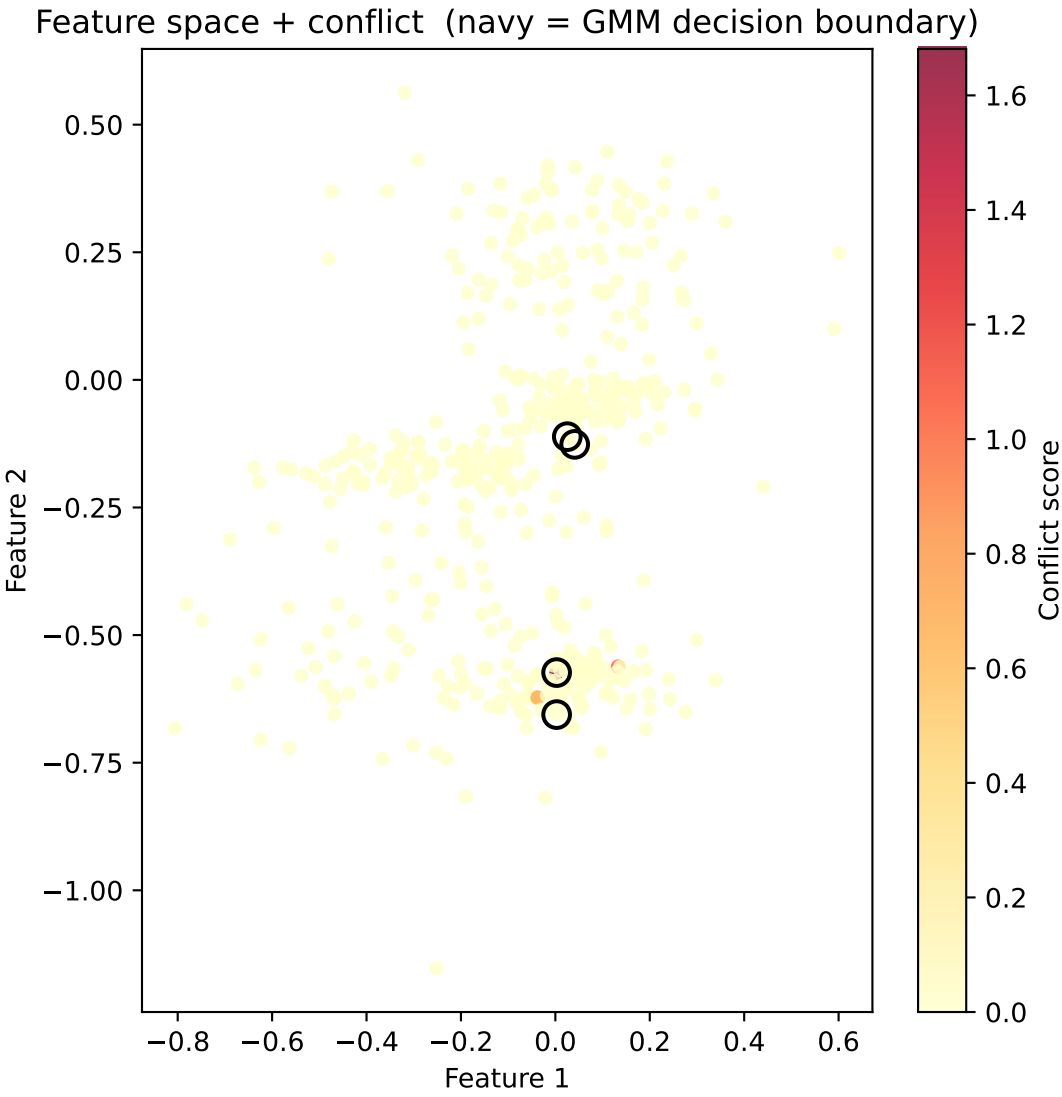
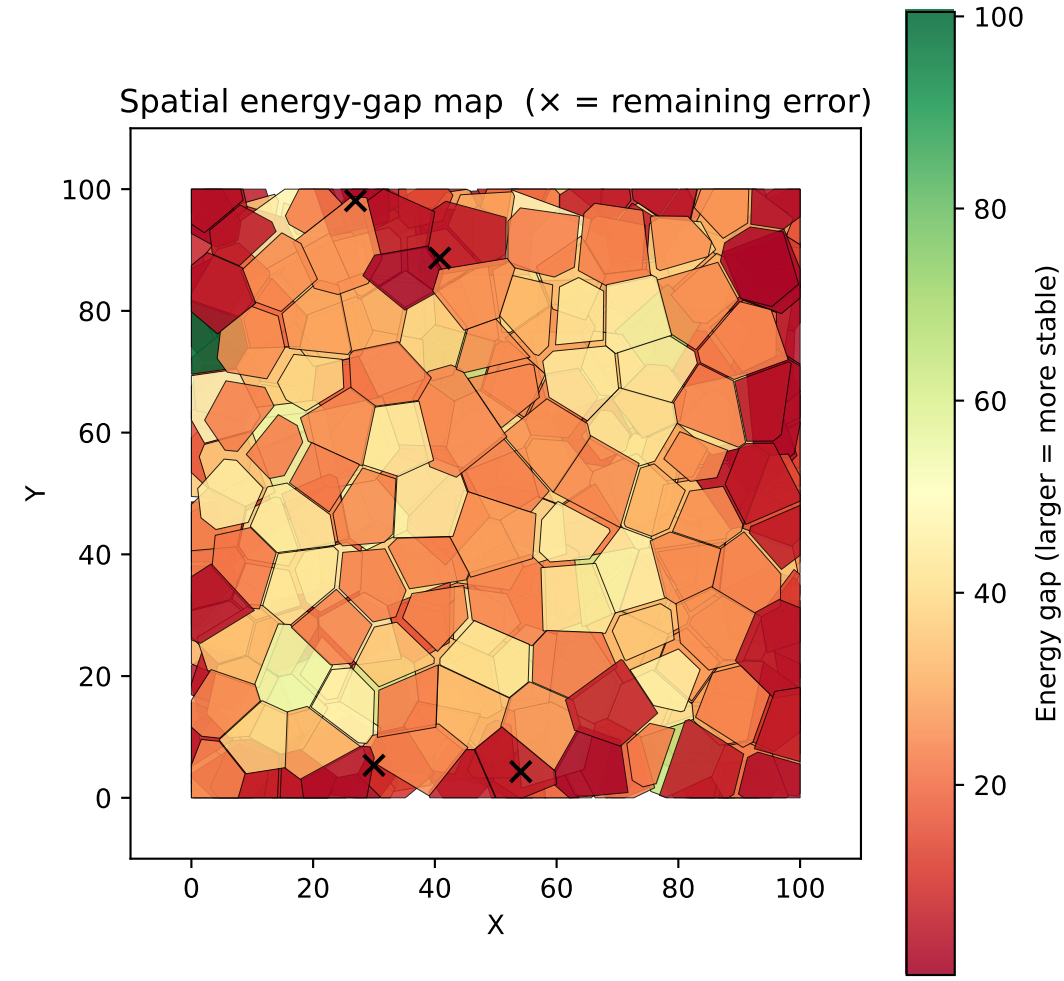
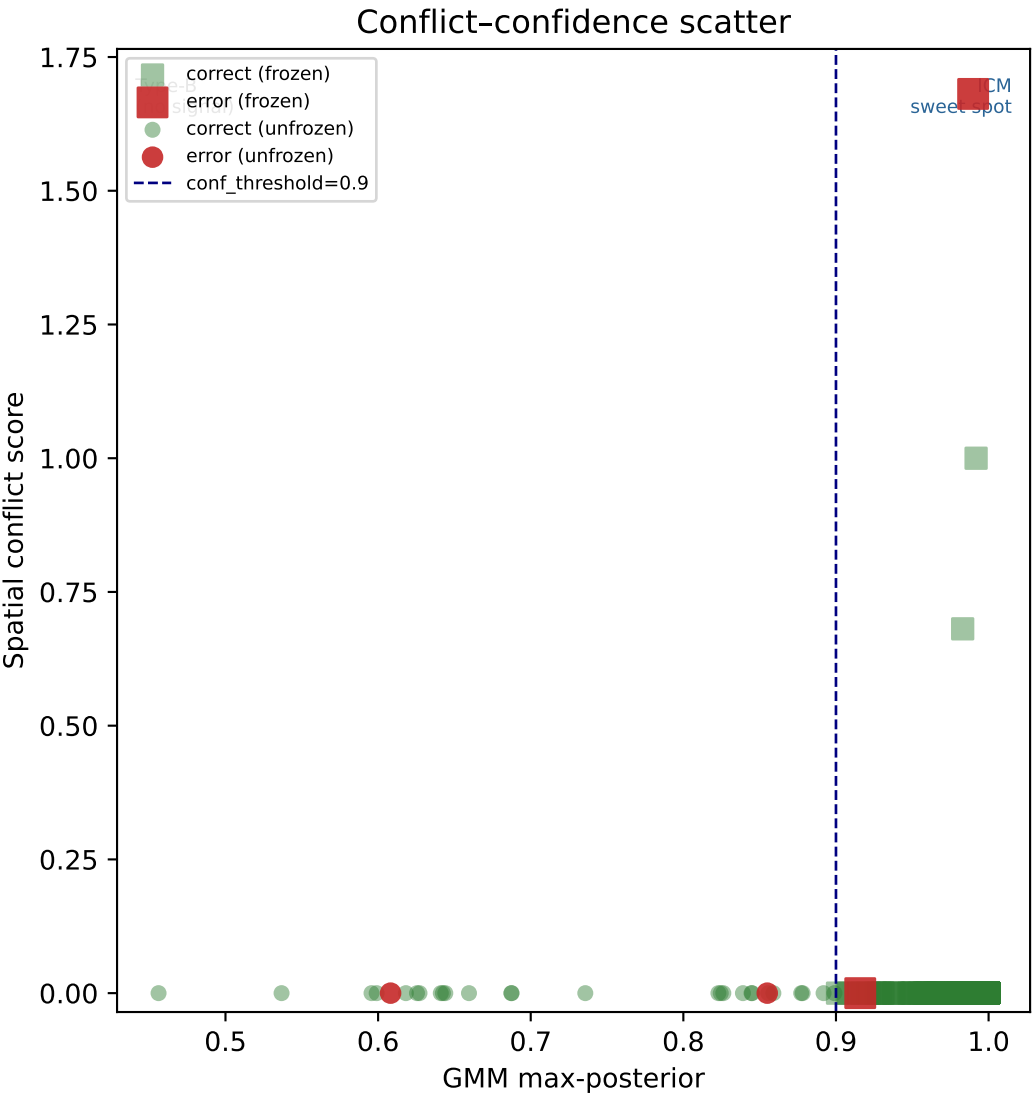
| MRF Run Summary |                   |
|-----------------|-------------------|
| ARI:            | 0.9799            |
| errors:         | 4 / 490           |
| violations:     | 16 → 2            |
| n_changed:      | 10                |
| n_iters:        | 2                 |
| n_frozen:       | 462               |
| Parameters:     |                   |
| lam:            | 20.0              |
| tau_low:        | 0.3               |
| tau_high:       | 0.4               |
| radius:         | 7.059097824976937 |
| conf_thr:       | 0.9               |

elongated — GMM + MRF (signed log-ramp  $\alpha=10$ ,  $\tau_{\text{low}}=0.3$ ) ARI=0.980 violations 16 -> 2 changed=10 iters=2





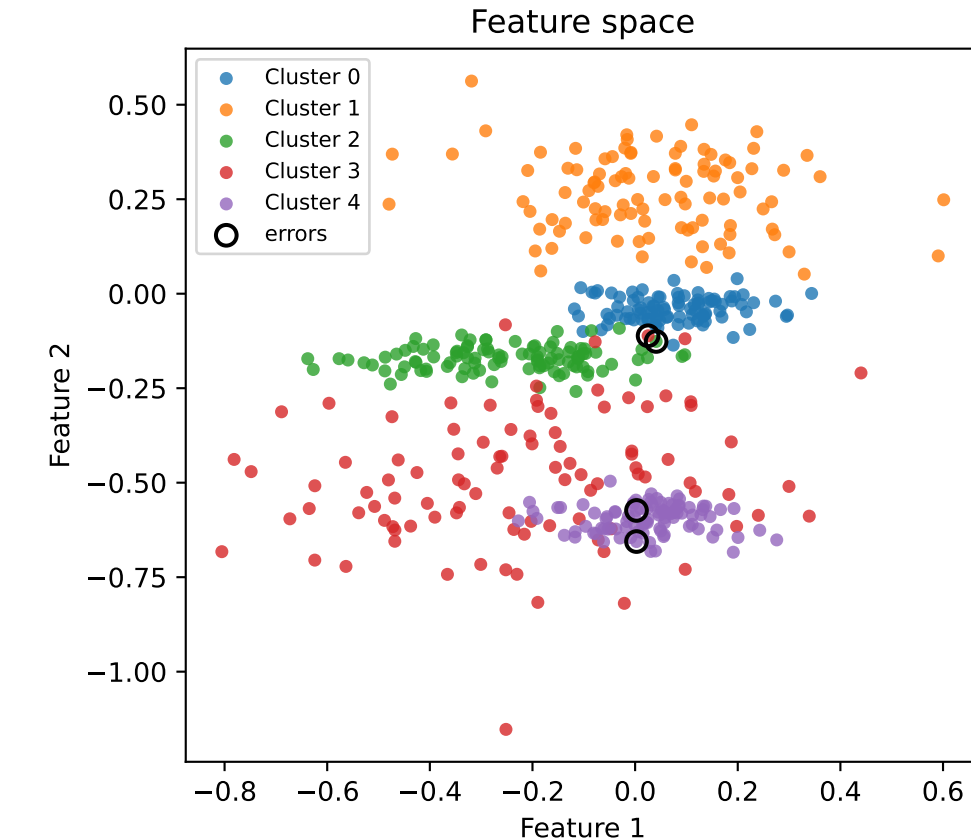
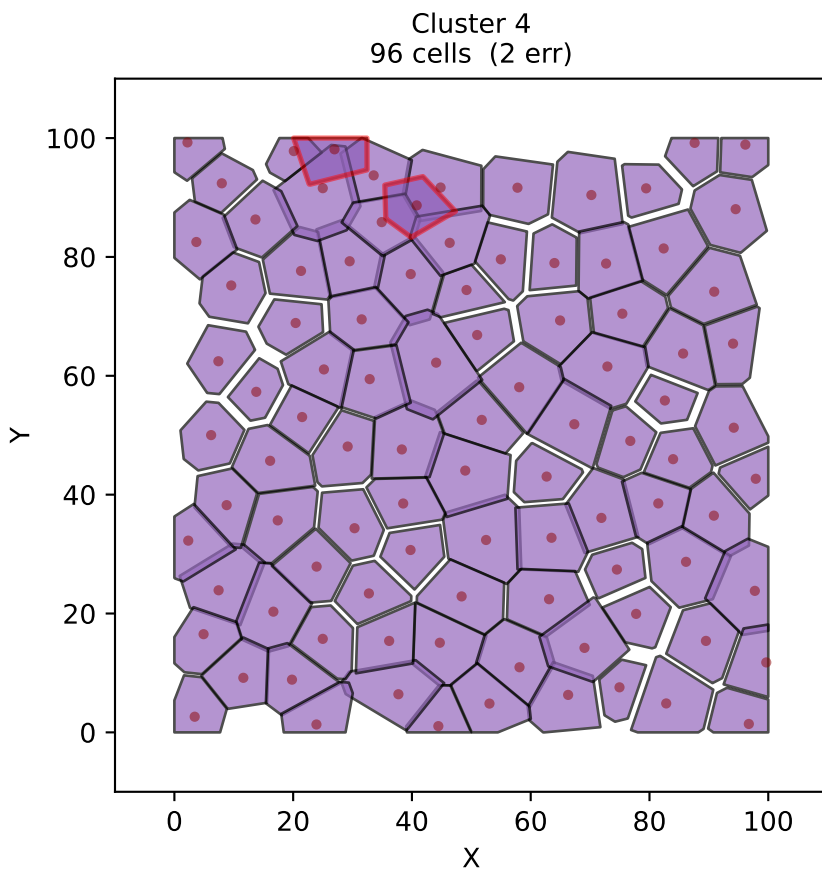
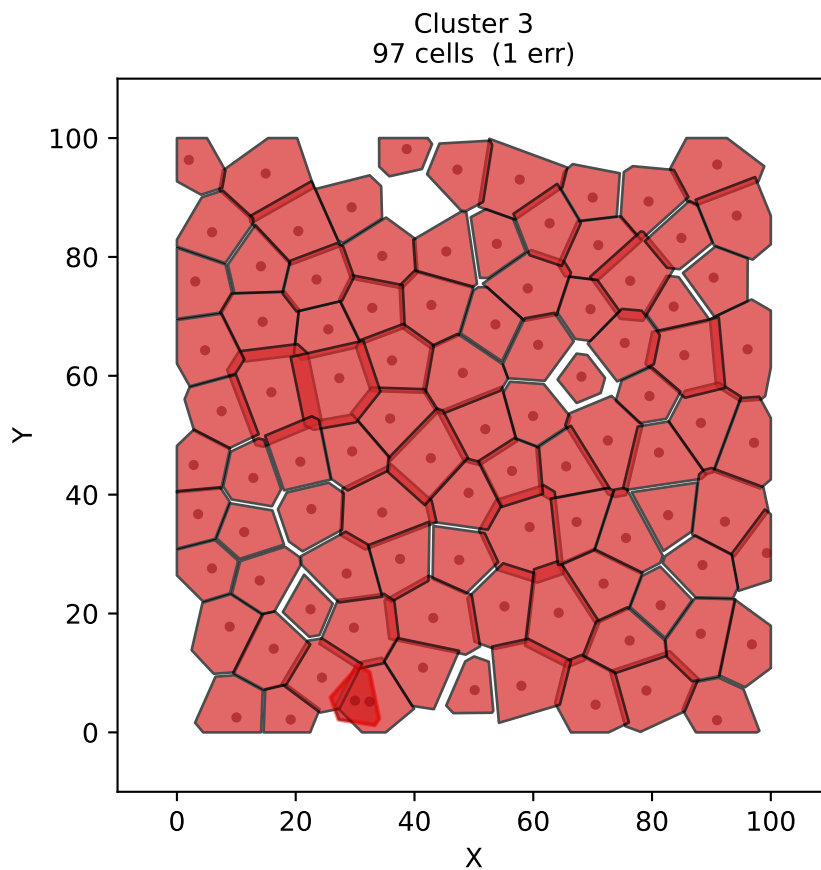
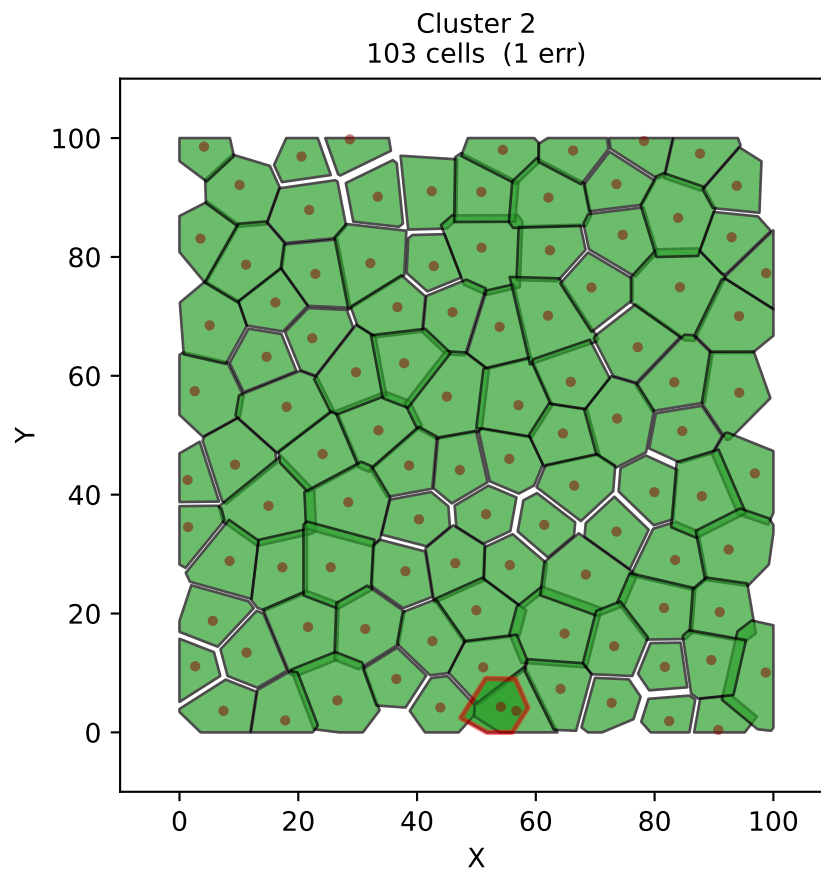
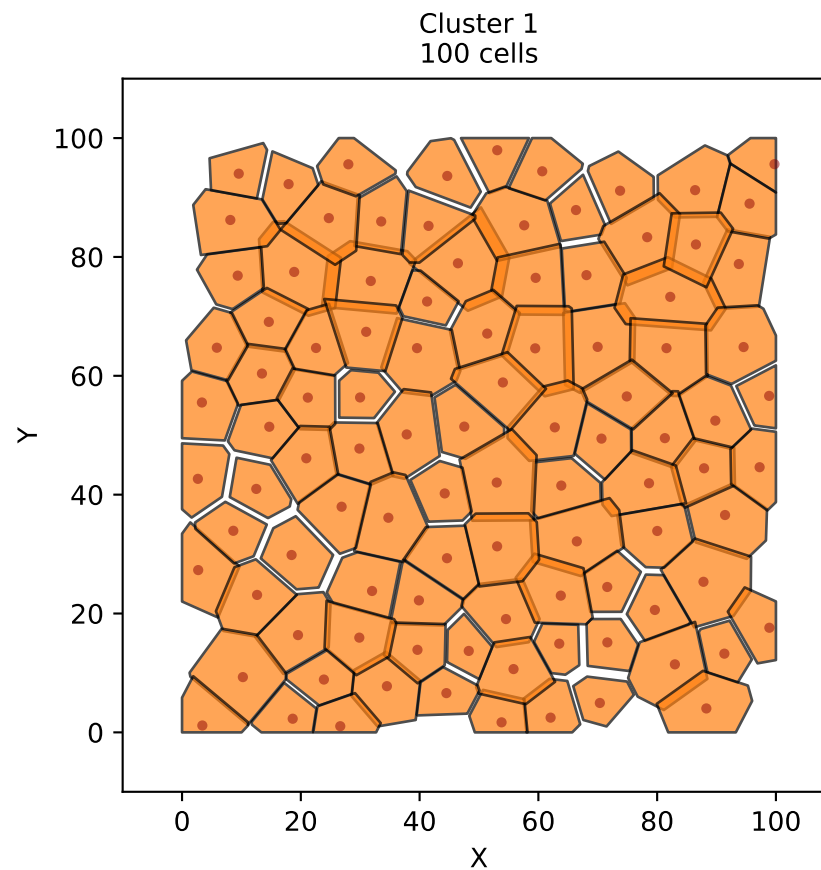
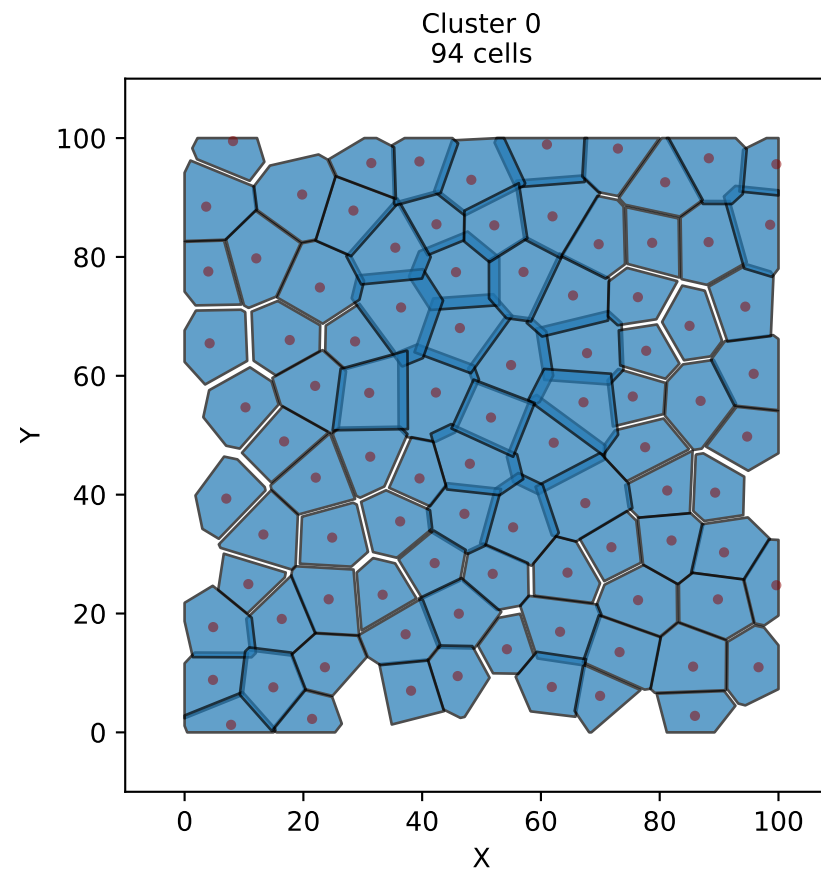
elongated — MRF diagnostics (signed log-ramp  $\alpha=10$ )



Run summary

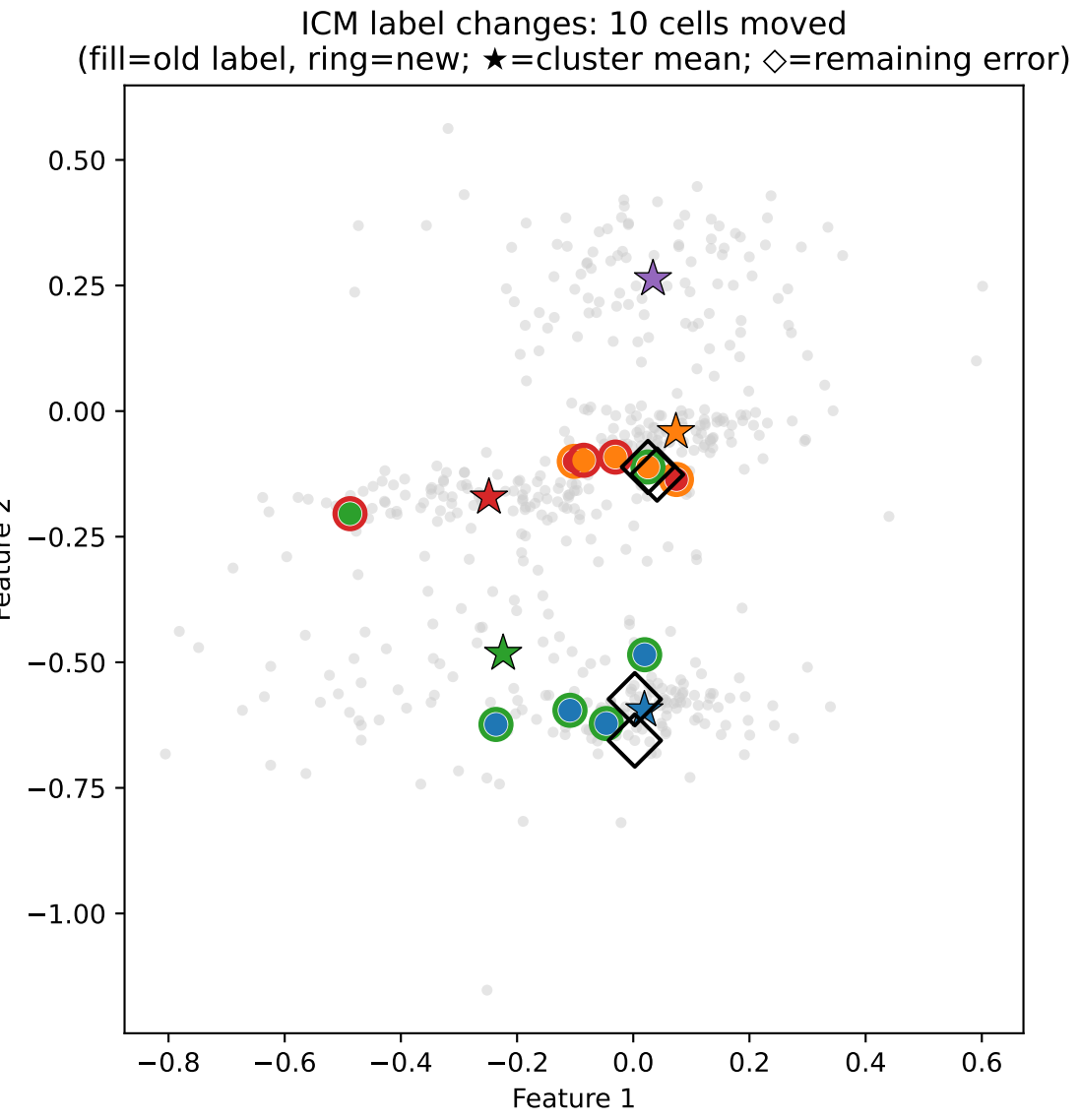
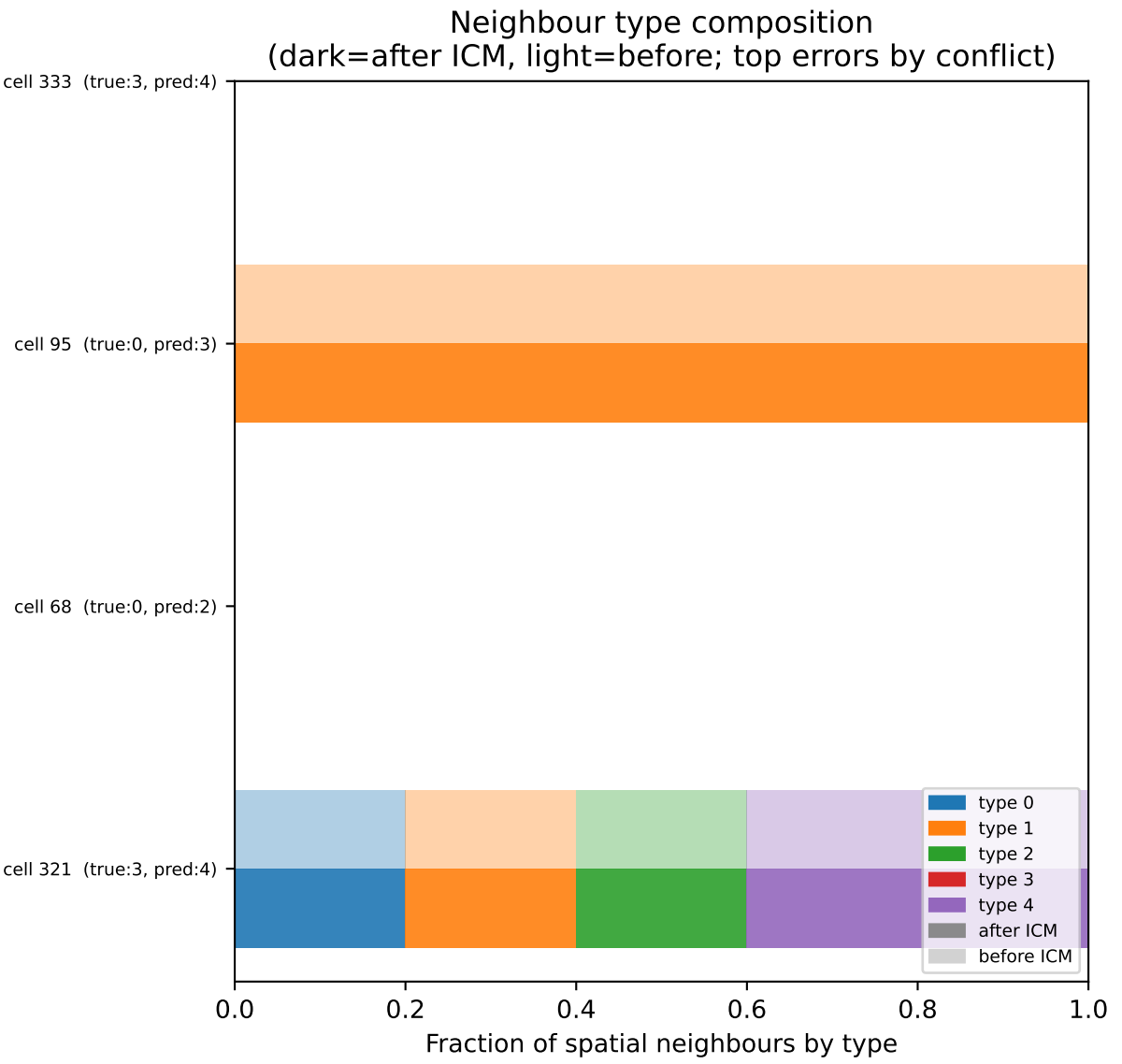
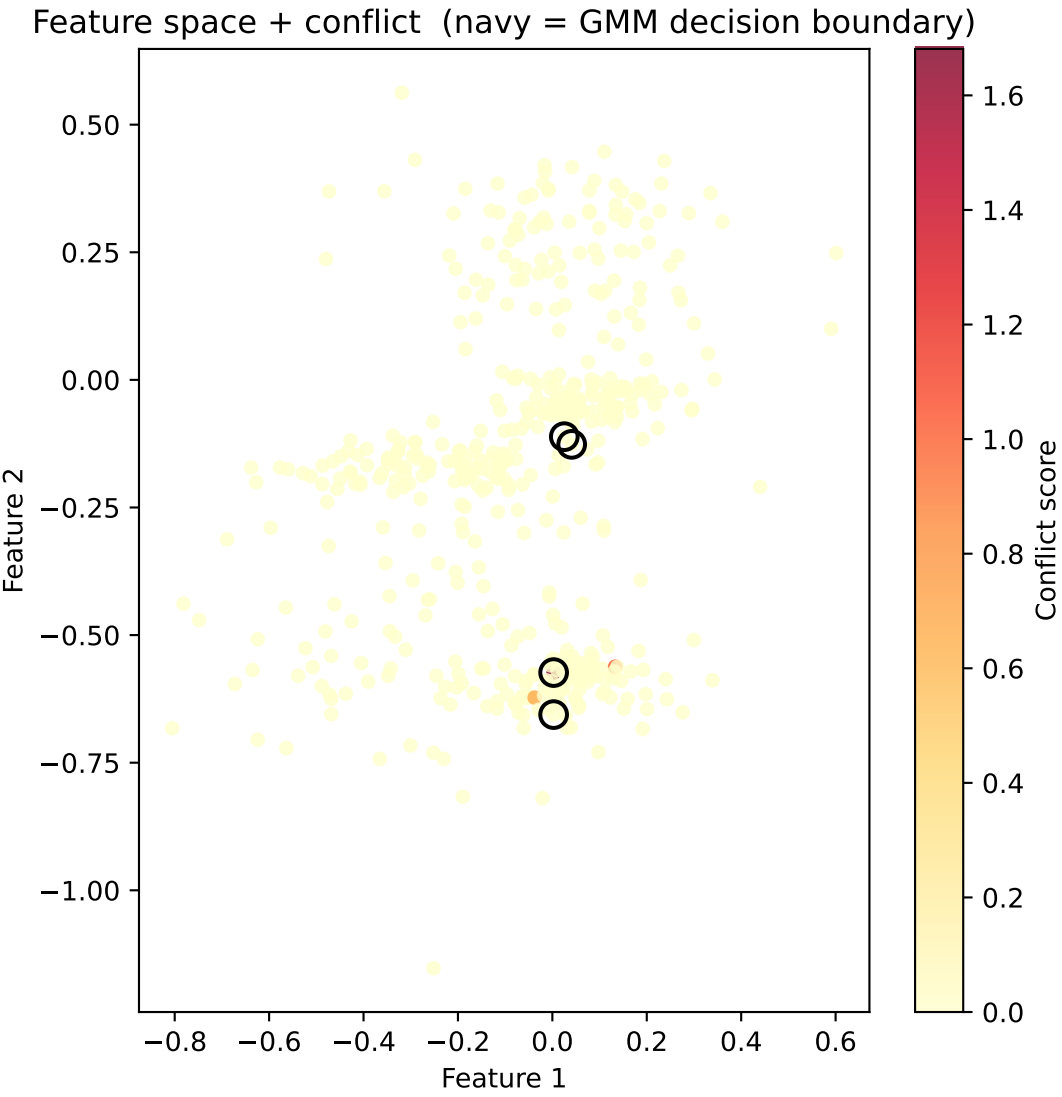
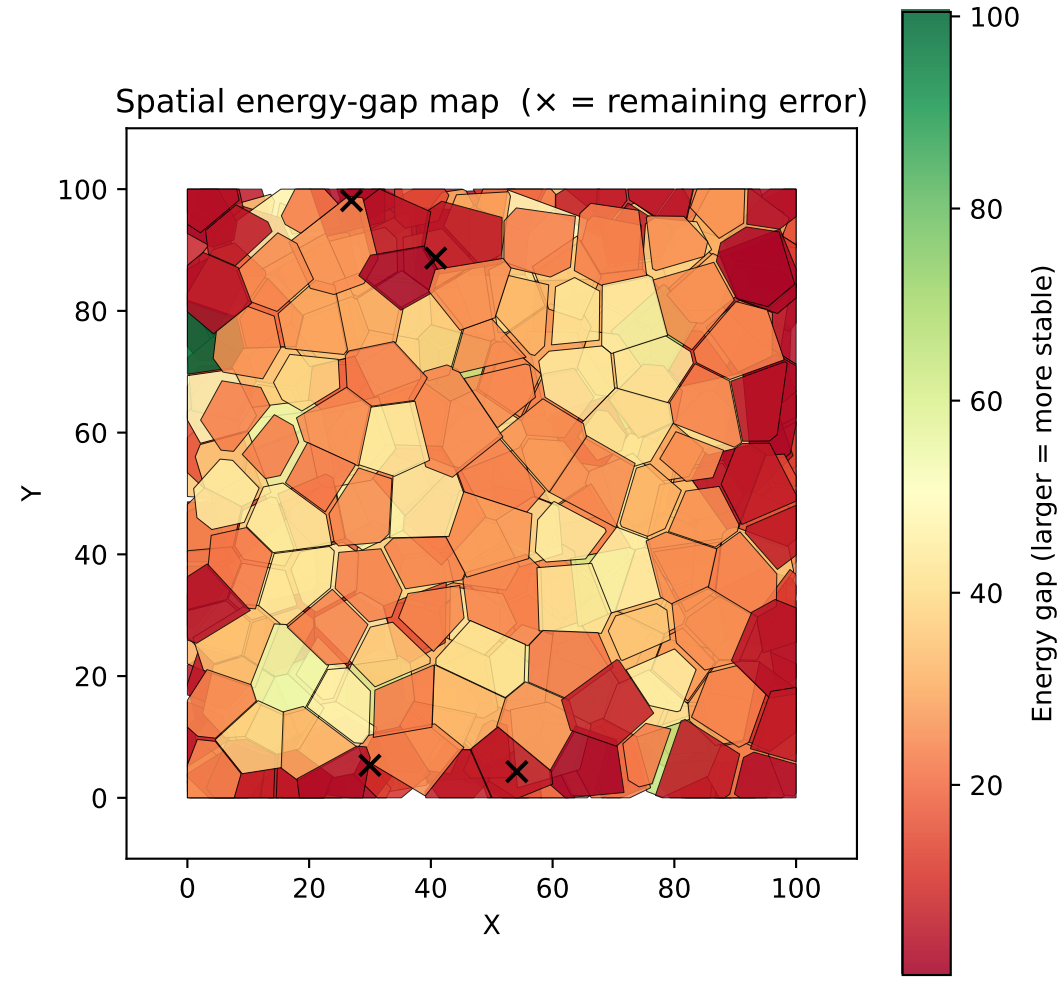
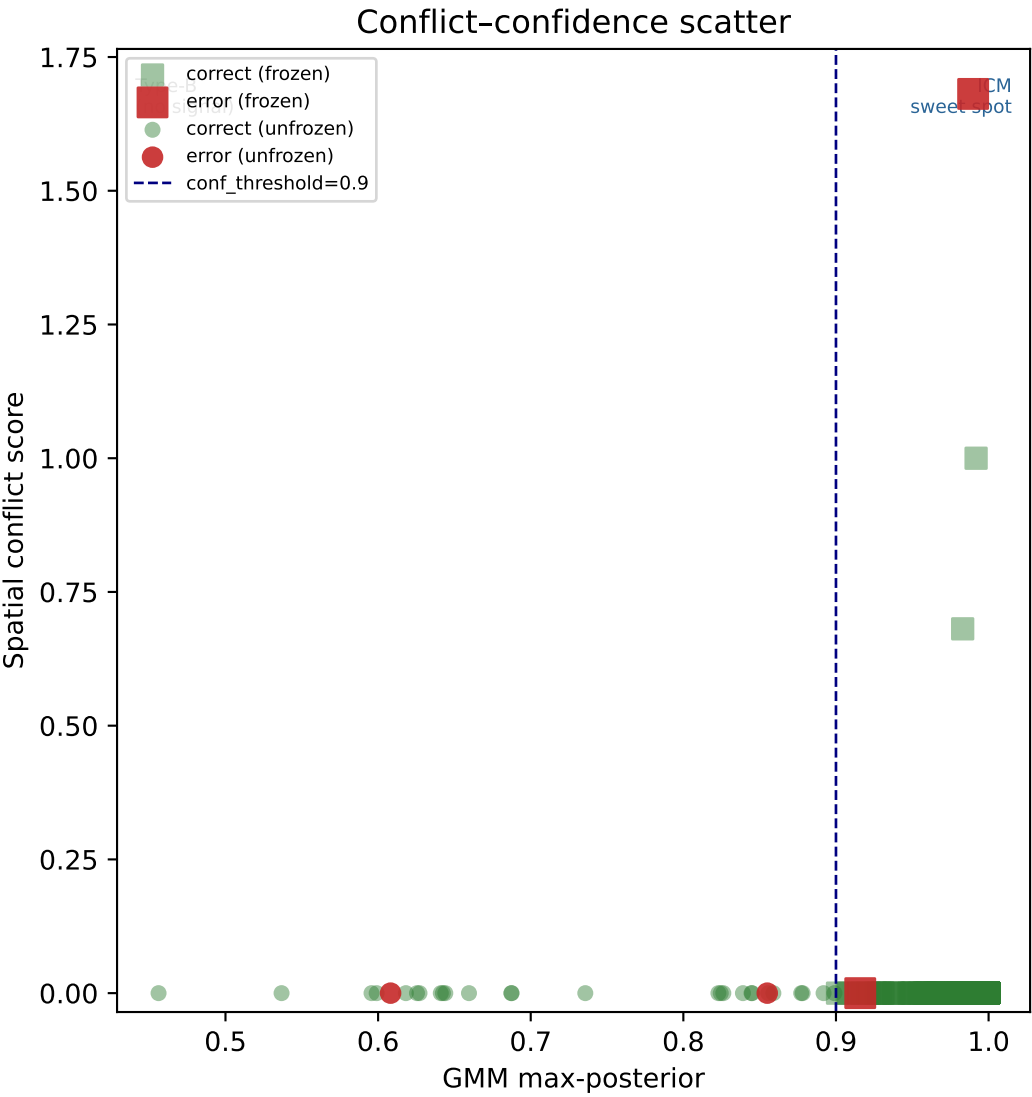
| MRF Run Summary |                   |
|-----------------|-------------------|
| ARI:            | 0.9799            |
| errors:         | 4 / 490           |
| violations:     | 16 → 2            |
| n_changed:      | 10                |
| n_iters:        | 2                 |
| n_frozen:       | 462               |
| Parameters:     |                   |
| lam:            | 20.0              |
| tau_low:        | 0.3               |
| tau_high:       | 0.4               |
| radius:         | 7.059097824976937 |
| conf_thr:       | 0.9               |

elongated — GMM + MRF (signed log-ramp a=10 + H1 th=0.75) ARI=0.980 violations 16 -> 2 changed=10 iters=2 force\_unfrozen=0





elongated — MRF diagnostics (signed log-ramp + H1)



Run summary

| MRF Run Summary |                   |
|-----------------|-------------------|
| ARI:            | 0.9799            |
| errors:         | 4 / 490           |
| violations:     | 16 → 2            |
| n_changed:      | 10                |
| n_iters:        | 2                 |
| n_frozen:       | 462               |
| Parameters:     |                   |
| lam:            | 20.0              |
| tau_low:        | 0.3               |
| tau_high:       | 0.4               |
| radius:         | 7.059097824976937 |
| conf_thr:       | 0.9               |